

Ion exchange process: History, evolution and applications

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Summary. — The aim of this paper is to present a review on some aspects and applications of ion exchange process in glasses, ferroelectric and polymers in the fields of optics, nanotechnology, gas sensors and chemical strengthening. The formation of nanoparticles in ion-exchanged glasses, as effect of ion or laser irradiation, is discussed. A discussion on the potentialities of ion exchange process in comparison to ion implantation in optical devices and nanotechnology is also introduced. Analytical techniques applied to the study of the ion exchange process are illustrated. The studies of ion exchange process in “Natural materials” constitute the content of a specific paragraph, for applications in water cleaning. Some initial considerations on the “old age” of this technique are introduced.

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1. – The “old age” of ion exchange

The ion exchange technique is very old! The first example, to our knowledge, is in the field of cultural heritage. Luster decorations, typical of Medieval and Renaissance pottery of the Mediterranean basin, substantially consist of a glass thin film, composed of a heterogeneous distribution of silver and copper nanoparticles of sizes ranging from 5 to 100 nm (*i.e.* of a metal-glass nanocomposite thin layer [1-4]), deposited on ceramic substrate. The decoration is painted by depositing a mixture of copper and silver salts and oxides, together with vinegar, ochre, and clay, on the surface of a previously glazed pottery. Then the whole system was heated to about 600 °C in a reducing atmosphere produced by the introduction of smoking substances in the kiln. In these conditions, an ion exchange process (!) between metal and alkali ions in the glass was induced and a subsequent diffusion of such ions inside the glass. Later on, the metal ions, followed by a metal ion reduction, aggregated, remaining trapped within the first layer of the glaze. These decorations show peculiar optical properties, producing brilliant reflections of different colors, shot effects and iridescence, as shown in fig. 1. Luster preparation is described by Cipriano Piccolpasso in the second book of “I Tre libri dell’arte del vasaio”, 1557. The formation of nanoparticles (see the inset of fig. 1) is strongly analogous to that of the modern synthesis of metal-glass nanocomposites, consisting of ion exchange process followed by annealing in reducing atmosphere or low-mass ion irradiation [5-8]. Historically, luster was developed in Mesopotamia during the IX century, in Bagdad, Samarra, near the Tigris River, the second imperial residence of Abbassides. Subsequently we found lusters at Fustat-al-Qatai in the district of Misr (Egypt). This technique subsequently diffused throughout the Mediterranean basin together with the spread of the Islamic culture. It arrived at Spain during the XII century and was particularly developed in Paterna and Manises, near Valencia, during the XIV and XV centuries. In fig. 2 we report a blue cobalt and metallic luster monochrome on white substrate, representing the coat of arms of Ricci Family from Florence, produced in Manises (1470).

The starting of metallic lustre in Valence gave to this technique a new commercialization area, with a developing interest in Italy, in particular in Tuscany and in the centre of Italy, where it was exploited to produce the well-known polychrome luster Renaissance pottery of Deruta and Gubbio. This nanotechnology has been transferred from



Fig. 1. – Lustre prepared by Mastro Giorgio in Gubbio (Italy, 1528) (Museo Civico della Ceramica di Gubbio, Italy). The inset is a TEM image of copper nanoclusters in the gold lustre.

Mesopotamia to Italy in 700 years! Moreover “Nature” used negatively the ion exchange process in the corrosion of beautiful medieval cathedral windows in several European countries. Such damage is more evident on glasses containing alkali elements as potassium at concentration higher than 14% instead of sodium, because of the easier exchange



Fig. 2. – Manises, 1470 blue cobalt and metallic luster monochrome on white substrate. Coat of arms of Ricci Family from Florence.

process between proton and potassium. The consequence is the formation of craters on the glass. The knowledge of surface degradation of historical glasses, exposed for centuries to weathering, is very important. It is really interesting to understand the type and features of the chemical-physical phenomena which occur on these materials, like change in surface composition, chemical stability, different ways of leaching and alteration products. Particular interest assumes the study of the alteration processes that occur on glasses immersed in seawater.

The most important reactions that can take place when silicate glasses are in contact with an aqueous environment at $pH < 9$ are ionic exchange reactions between alkali ions of glass and protons of the solution [9], while aqueous solutions at $pH > 9$ induce the destruction of the bridging oxygen bond of the silica network [10]. The substitution of alkaline or alkaline-earth ions with protons also causes a volume contraction of the vitreous matrix that induces fracture, collapse of the structure and pit formation. In pits crystallization causes a break-up action similar to the one of ice inside rock fractures. An interesting study [11] was performed on glass fragments coming from the Roman ship Iulia Felix, which sank near Grado (Gorizia Italy) in the second century A.D. [12]. The glass composition is typical of the Roman era, with a large use of “natron” (“natron” is a melting agent, $Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O$) and with small quantities of Mediterranean and continental plant ashes [12-14]. Detailed XPS and SIMS analyses have shown that the evolution of the glass submerged in marine water is similar to the one found in glasses from other terrestrial archaeological sites. The alteration processes, like ion migration, formation of calcium accumulation layers and carbonates near the surface, are mostly similar to the ones observed in other Roman and early medieval glasses. The only difference is that when glass fragments remain for several centuries submerged in marine water, all the processes are faster and the final detectable results are more evident. So the presence of carbonates or hydrated layers inside the glass is very relevant and it produces a strongly visible exfoliation layer.

The ion exchange scientific and technical application [15-17], however, is dated back to 100 years ago when it was first applied in the chemical surface tempering of glasses (potassium ion exchange).

2. – Ion exchange in glass

2.1. Ion exchange techniques. – Ion exchange in silicate glasses is known since 1913 [18] and it has been formerly studied for chemical strengthening of glass surface [19]. To date it is studied for the fabrication of passive and active optical waveguides [20-23], of stress-modulated systems [24], for the control of microstructural properties of glasses [25] and for deepening the understanding of environment-glass interaction mechanisms [26].

The most used procedure for ion exchange in glass is based on the immersion of glass slides in mixtures of molten nitrate salt baths, at temperatures lower than the glass transition temperature. In order to tailor the concentration profile of the diffusing ions, different approaches were also tested: electric-field-assisted diffusion from a molten salt bath was used for the production of step-like diffusion profiles [27,28], electric-field-assisted ion diffusion from a thin metal film deposited on the glass surface was used as alternative method also for supplying ions which are not available as low melting salts [29-32] and the combination of the two above-described procedures in order to design buried profiles with a proper shape [33-36].

Up to now, this process has been performed mostly by substituting alkali ions in silicate glasses. In particular the most substituted ion was Na^+ , in glasses where the

Na_2O molar concentration was higher than 10%, so as to let a sufficiently high amount of doping ions enter the glass, and thus having the possibility to tailor the surface physical properties. Sometimes, for introducing larger ions like Cs^+ , potassium-rich silicate glasses were treated, since K^+ provides larger sites for the accommodation of the incoming ion species. For the development of industrial optoelectronic devices, as will be explained in the next section, special glasses were produced to overcome the drawbacks arising for the ion exchange treatment. Substitutive ions were in most cases K^+ , for surface strengthening and optical purposes, and Ag^+ for the production of optical waveguides. In few cases other alkali ions have been used, like Li^+ [37] or Cs^+ [38, 39], and metal ions like $\text{Cu}^+/\text{Cu}^{2+}$ [40-43], Tl^+ [44], Er^{3+} [45] and more recently Co^{2+} , Au^{3+} and Cr^{3+} [31, 32].

The laboratory equipment for processes based on thermal diffusion and exchange is very simple, since it is constituted by an oven with a suitable container for the salt melt and the accurate control of the process temperature is the only key factor for reproducibility of the process. On the other hand, the processing equipment for field-assisted processes with molten salts is more complex, as electrical contacts must be made to the anode and cathode sides of the substrate, and these need to be maintained in electrical isolation. The literature on waveguide fabrication gives many descriptions of process arrangements; a more detailed description of the equipment for field-assisted ion exchange can be found in Monteiro *et al.* [46].

Due to the operational setup difficulties when dealing with molten baths, most of the field-assisted experiments were realized by promoting the ion migration into the glass from a metallic solid film deposited on the surface with electric contacts which were usually Al or Au thin films or molten salts. The external electric field induces the oxidation of the metallic film and the migration of ions in the glass network, whereas Na^+ migrates to the opposite side. The great advantage of this procedure is to largely extend the amount of ion species which can be introduced into the glass network, overcoming the limitations related to the availability of nitrates, which are salts melting at low temperature with respect to the softening temperatures of the employed glasses. From the first experiments with metallic films as ion sources the most used element was silver, due to its wide application in the fabrication of passive optical devices and to the possibility of producing buried waveguides, by promoting the exhaustion of the whole film. After the first works on the production of Ag^+ -containing optical waveguides from thin metallic films, during the last years the diffusion from solid-state sources was used for producing samples with Cu^+ [42, 47, 48], Co^{2+} , Au^{3+} , Er^{3+} and Cr^{3+} [31, 32, 49, 50].

In spite of the intriguing possibilities opened by ion migration from thin films, several technical problems are still open. At first, space charge can occur near the ion-injecting anode if the diffusion rate of the invasive ions is not equal to the rate at which the glass ions move to the cathode. The same problem can occur if the cathode cannot remove or neutralize the glass ions at the required rate. Moreover, the formation and diffusion of ions at the glass-film interface is a critical issue in controlling the process rate and the homogeneity of the sample. Up to now, few papers deal with the behavior of the glass-film interface during the field-assisted migration from a thin film. Among them, the deeper analysis has been presented by Kapila and Plawsky [51, 52], who assumed the formation of an oxide buffer layer as ion source for the glass, describing the mechanism with the electrochemical cell hypothesis and describing the current behavior in terms of the film consumption. More recently, it was demonstrated that the migration of Cr^{3+} in soda lime or borosilicate glass can be thermally activated from an oxidized film deposited on the surface, opening new possibilities to the doping of glasses with multivalent metal

ions. In this case simple heating of the sample at 400–500 °C promoted the diffusion of a Cr^{3+} -rich phase up to several microns beneath the glass surface [53].

2.2. Ion exchange modeling. – In order to describe the properties of ion-exchanged glass, it is necessary to model the concentration profiles of the exchanging ions as a function of the processing conditions (temperature, exchange time, applied electric field, etc.).

Concerning the salt bath ion exchange, at the glass surface the following chemical balance occurs:



where A^+ is the alkali ion of the glass and B^{z+} is the doping ion from the salt and z is the oxidation state of the incoming ion, which determines how many alkali ions have to exchange for compensating the charge. The exchange at the surface is driven by the difference chemical potentials in the two phases and it occurs much faster than the mass transport in the glass. For this reason, in the analysis of the inter-diffusion process the surface concentration can be considered as a constant. So, the chemical equilibrium at the glass-melt interface affects the ion concentration and, for low concentrations, the shape of the diffusion profile.

The equilibrium conditions at the glass surface have been studied by Schulze and Garfinkel [18, 54] and by Houde-Walter and Moore [55] for different cation pairs and glass compositions. One of the parameters involved both in the surface equilibrium conditions and in the ion diffusion in glass is

$$(2) \quad \frac{\partial \ln a_A}{\partial \ln X_A},$$

where a_A and X_A are, respectively, the thermodynamic activity and the fraction of ion A in the glass. This term is usually considered to be constant, following for instance the glass-salt equilibrium studies of Garfinkel [54], who interpolated the experimental data by supposing that the logarithmic derivative is equal to a constant n between 1 and 2. More recently, Araujo found an expression, obtained from the statistical thermodynamics relationship of the electrochemical potentials, for evaluating the interaction energy between unlike ions from glass-salt equilibrium curves [56].

In the glass network, ionic diffusion is governed by the Nernst-Planck equations for the ion fluxes J_i of the migrating species

$$(3) \quad J_i = -D_i \frac{\partial \ln a_i}{\partial \ln X_i} \frac{\partial c_i}{\partial x} + \mu_i E c_i,$$

where D_i is the self-diffusion coefficient of the ion species, a_i the thermodynamic activity, X_i and c_i the atomic fraction and the concentration, respectively, E the local electric field balancing the local charge density when two ions with different mobility are diffusing, and μ_i is the ion mobility. The latter parameter is related to the diffusion coefficient by the Nernst-Einstein relation

$$(4) \quad D_i = f_i \frac{kT}{z_i e} \mu_i,$$

where the factor f_i accounts for the difference between thermal and field-assisted migration, k is the Boltzmann constant, T the absolute temperature, z_i the ion valency, and e is the elementary charge (for an exhaustive description of the term f_i , see refs. [57,58]). Self-diffusion coefficients follow an Arrhenius dependence on the temperature

$$(5) \quad D_i = D_{0i} \exp(-\Delta E/kT),$$

where ΔE is the energy barrier for a jump between two adjacent sites and D_{0i} is the pre-exponential factor. The first attempt to derive an expression for the energy barrier is due to Anderson and Stuart [59], which assumed that it was constituted by two terms, related to the electrostatic interaction with the neighbors and to the elastic strain of the network, treated as a continuum.

The final differential equation for the concentration of the ion is obtained by removing the local electric field term, under the conditions that the local charge neutrality is attained

$$(6) \quad Z_A J_A + Z_B J_B = 0, \quad Z_A J_A + Z_B J_B = Z_A c_{0A},$$

where c_{0A} is the pristine concentration of A^+ ions in the glass. In literature [60,61], the correctness of this assumption was verified by explicitly calculating the local electric by numerical methods during the ion exchange, finding that the local charge neutrality is satisfied within 10^{-3} s from the starting of the inter-diffusion process.

By combining eqs. (3) and (6) with the condition on the thermodynamic activities for the local equilibrium

$$(7) \quad \frac{\partial \ln a_A}{\partial \ln X_A} = \frac{\partial \ln a_B}{\partial \ln X_B}$$

the differential equation for the concentration profile $c_B(x, t)$ of a monovalent ion takes the form

$$(8) \quad \frac{\partial c_B}{\partial t} = -\frac{\partial J_B}{\partial x} = \frac{\partial}{\partial x} \left(\frac{\partial \ln a_B}{\partial \ln X_B} \frac{D_B D_A}{X_A D_A + X_B D_B} \frac{\partial c_B}{\partial x} \right) = \frac{\partial}{\partial x} \left(\tilde{D} \frac{\partial c_B}{\partial x} \right),$$

where $\tilde{D}$ is the inter-diffusion coefficient.

This relationship, firstly introduced by Doremus [15], is the equation used for the analysis of the diffusion process of the doping ion in the glass matrix. The term containing the thermodynamic activity is usually neglected or kept constant, following for instance the work of Garfinkel [54]. So, the equation can be solved numerically or by means of analytical procedures, by taking into account the dependence on the concentration of the inter-diffusion coefficient.

Some limiting cases can be solved analytically, for example when $D_A = D_B$ or when $X_A \ll 1$, the latter occurring in very dilute salt solutions. Under these conditions the inter-diffusion coefficient is simply equal to D_B and the equation can be solved analytically by assuming a constant concentration at the surface c_{0B}

$$(9) \quad C_B(x, t) = C_{0B} \operatorname{erfc} \left(\frac{x}{2\sqrt{D_B t}} \right).$$

In general, the diffusion profile resembles a Gaussian shape, while for $D_B \ll D_A$, like in the case of Cs^+ - Na^+ exchange [38], it takes the form of a step-like profile.

When an external field is applied, in eq. (3) the electric field is the sum of the local field and the external field and the sum of the two ion fluxes is equal to the total current flux through the glass sample. In this case the flux equations for the two ions combined with the conditions on both local charge balance and charge fluxes have to be solved by means of numerical methods. Only for dilute systems, when the inter-diffusion coefficient is equal to the incoming ion self-diffusion coefficient, and when the gradient of the ion concentration is 0 both at the surface and at the end of the sample, an analytical solution can be obtained by the Laplace-transform technique

$$(10) \quad C_B(x, t) = \frac{c_{0B}}{2} \left\{ \operatorname{erfc} \left(\frac{x - \mu_B E_{\text{ext}} t}{2\sqrt{D_B t}} \right) + \exp \left(\frac{x \mu_B E_{\text{ext}}}{D_B} \right) \operatorname{erfc} \left(\frac{x + \mu_B E_{\text{ext}} t}{2\sqrt{D_B t}} \right) \right\}.$$

For large electric fields it can be shown that the second term in eq. (10) can be neglected and diffusion does not play a significant role giving a step-like profile. From the analytical point of view, the problem has been studied by Abou-el-eil and Cooper [62] and then by Oven and co-workers [35, 63, 64], giving several approximate solutions both for single and multiple ion exchange treatments.

2.3. Ion exchange effects on the glass structure. – For several years, in literature ion exchange was considered a mere insertion of a new ion in the Na^+ site, with the possible formation of only stress states given by the difference between the ionic radii of the exchanging species. With the rising of experimental techniques, such as extended X-ray absorption fine structure (EXAFS) spectroscopy, the knowledge of the short-range glass structure reached new aims. In 1981 Greaves *et al.* [65] found for the first time that in sodium silicate and soda-lime glasses the coordination of sodium follows a short-range order similar to crystalline silicates, with coordination numbers (CNs) of about 5 for sodium silicate and of about 2 for soda-lime glasses, with Na-O bond lengths in the range of 0.220–0.240 nm. Moreover, in glasses usually employed for ion exchange experiments, the sodium concentration is so high to form percolation channels in the silica network. This frame, which strongly affects the diffusion mechanism, can change in a more random distribution for the presence of Al_2O_3 , which lowers the number of non-bridging oxygen (NBOs).

Concerning ion exchange, a good deal of work has been devoted to EXAFS analyses on silicate glasses treated by Ag^+ - Na^+ ion exchange. In particular, both Dubiel *et al.* [66, 67] and D’Acapito *et al.* [68] found that the ionic site around Ag^+ ions is rearranged owing to the formation of nearly covalent bonds with NBOs, leading to the lowering of the Ag-O distance to about 0.215 nm and to the change of the CN from 5 to 3 for low Ag^+ concentrations (12% of exchange) and to 2 for high concentrations (80% of exchange). In particular, the latter structure resembles that of Ag_2O , even if the bond length of this oxide is lower (about 0.204 nm). It is worth noting that the Ag-O bond length has been well reproduced in two different types of glasses. In addition, experimental findings combined with molecular dynamics calculations [69] established that the Ag-O bond length has a slight tendency to decrease with the increasing of the silver content in the silicate structure, developing a Ag-Ag correlation attributed to homoatomic interactions between closed d^{10} shells.

More recently, it has been found that the Ag-O bond length for low exchange times (few minutes) is higher [70, 71] with respect to what found by Dubiel, who examined glass

samples treated for very long times (several hours). This result evidenced a structural relaxation behavior similar to that observed for the K^+ - Na^+ ion exchange, where the surface stress state tends to relax by increasing the treatment temperature and time [72]. By combining Raman spectroscopy with EXAFS measurements, a progressive depolymerization of the silica network with the increase of non-bridging oxygens was evidenced, with the increase of silver content. The depolymerization is accompanied by the aggregation of silver in metal nanoparticles, which can be evidenced by the increase of the acoustic resonance peak in the low-frequency region of the Raman spectra but not by other spectroscopic techniques, like optical absorption [73].

The strong rearrangement of the glass short-range structure around Ag^+ explains the absence of relevant stress states, as observed by measuring the birefringence of optical waveguides [74, 75]. Indeed, the Ag^+ ionic radius (0.126 nm) is higher than Na^+ (0.095 nm), but does not give rise to measurable stress states in the glass network, as occurs for instance in the case of the K^+ - Na^+ ion exchange, where the K^+ ionic radius is 0.133 nm.

Even stronger effects on the glass structure have been evidenced by EXAFS measurements in Cu^+ - Na^+ exchanged soda-lime glasses [76, 41]. In this case, the doping ion diffused into the glass with two oxidation states, Cu^+ and Cu^{2+} , exploiting different coordination geometries. In particular, a square local symmetry around Cu^{2+} ions and a linear coordination of O atoms around Cu^+ have been evidenced. The bond lengths were characteristic of CuO (around 0.195 nm) where a higher amount of Cu^{2+} was present, and of Cu_2O (around 0.185 nm) where Cu^+ was the predominant species [41]. This strong effect on the structure can be also related to the high ion exchange temperatures (between 520 and 560 °C), required for melting copper salt eutectics, which promotes the local rearrangement of the glass structure to the more stable coordination.

All these results show that the ion exchange procedure heavily affects the glass microstructure, even if the treatment temperatures are well below the glass transition temperature of the substrate or, in other words, at temperatures at which the glass structure is usually assumed to be frozen. Hence, it is hard to approach the ion exchange model by assuming constant self-diffusion coefficients for the exchanging ions.

2.4. Mixed alkali effect in ion exchange. – The first detailed study about the dependence of self-diffusion coefficients on the concentration of the migrating species can be found in the works of Frischat and Varshneya on K^+ - Na^+ ion exchange [77, 78]. In particular, the inter-diffusion coefficient $\tilde{D}(c)$ was calculated with the Boltzmann-Matano method from the K^+ diffusion profile and it was found that the dependence on the concentration cannot be fitted with the expression in eq. (8), with constant self-diffusion coefficients. On the other hand, a better result was obtained by inserting the tabulated self-diffusion coefficients of mixed-alkali glasses at the different glass compositions encountered along the diffusion profile. The residual deviations have been reconciled with the strains produced around a cation jump site during ion exchange and, in turn, with the macroscopic stresses in the glass body. This result suggested that the mixed-alkali effect (MAE) could give a reasonable physical interpretation for the dependence of the self-diffusion coefficients on the concentration during the ion exchange.

MAE is one of the most widely studied phenomena in glass science. From the point of view of the ion diffusion process, it consists in the lowering of the self-diffusion coefficient of an alkali ion when another species is introduced into the network. Moreover, the self-diffusion coefficient curves of the two alkalis intersect at some particular alkali ratio where both cations have the same mobility. Even if a complete explanation of MAE is

still to come, it has been definitely established that it is related to the stabilization of different sites for the two alkali species, and to the formation of percolation pathways for the majority ion. This situation resembles that previously described for ion exchange.

In particular, the dependence of the migration parameters on the alkali concentration is related to the modification of the activation energy for the jump between two adjacent sites. In fact, the energy barrier is given by two terms, one accounting for the strain of the network doorway during the jump and another one accounting for the electrostatic interaction energy between neighboring ions and this term could explain the observed dependence on the cation concentration. These terms have been reported for the first time in 1954 by Anderson and Stuart [59] and, with some more sophisticated revision, their expression was substantially confirmed more recently [79, 80].

Concerning ion exchange, a model based on concentration-dependent diffusion activation energies is physically meaningful if it takes into account the local rearrangement occurring in the glass network during the process. From this point of view, the activation energy can be expressed in a more general way, like that proposed by other authors for MAE [81, 82].

Up to now, evaluations of the jump energy values have been attempted by Tomozawa, who made the first evaluations from mixing enthalpy values, and by Greaves, who connected conductivity and EXAFS measurements [83, 84]. Following a different approach, Araujo found an expression, obtained from the statistical thermodynamics relationship of the electrochemical potentials, for evaluating the interaction energy between unlike ions from glass-salt equilibrium curves [85].

All the listed approaches are based on equilibrium mixture models, even if it has been demonstrated that in most cases the ion-exchanged glass layers reach the equilibrium structure after very long treatment times. Nonetheless, from the point of view of the inter-diffusion equation a similar attempt can be developed by assuming that self-diffusion coefficients depend on the ion concentrations by means of concentration dependent activation energies, following the relationships

$$(11) \quad D_A = D_{0A} \exp \left(-\frac{x_A \varepsilon_{AA}}{kT} - \frac{x_B \varepsilon_{AB}}{kT} \right), \quad D_B = D_{0B} \exp \left(-\frac{x_B \varepsilon_{BB}}{kT} - \frac{x_A \varepsilon_{BA}}{kT} \right),$$

where ε_{ij} are the interaction energies between ion i and ion j and it can be assumed that $\varepsilon_{ij} = \varepsilon_{ji}$. An empirical exponential dependence on the concentration has been described also by Batchelor *et al.* [86].

The analysis of $\text{Cu}^+ \text{-Na}^+$ ion exchange was carried out by assuming the expression for self-diffusion coefficients described in eq. (11) by Gonella *et al.* [41, 87]. In this case, the presence of two Cu species diffusing into the glass under different oxidation states makes the description of the phenomenon much more complex. In particular, by integrating the coupled diffusion equations of the three migrating species, correlation effects between charge fluxes have been evidenced, giving rise to the establishing of a Cu^+ concentration profile with a maximum below the surface. This result was confirmed by the analysis of EXAFS spectra obtained at different depths, where the different $\text{Cu}^+/\text{Cu}^{2+}$ ratios have been calculated by interpolating the spectra assuming a mixture of CuO and Cu_2O structures. Finally, the experimental concentration profiles have been accurately reconstructed by expressing the self-diffusion coefficients with eq. (11).

For the sake of completeness, the approach carried on by Araujo [56, 88] and Houde-Walter and co-workers [89-92] has to be mentioned. From statistical thermodynamics, the authors found an expression for the dependence on the concentration of the logarithmic

derivative of the ion activity into the glass, while the self-diffusion coefficients were kept constant. The fitting parameters were the cation coordination, the ion-ion interaction energy and a phenomenological parameter. The inter-diffusion coefficient, calculated as a function of the ion concentration by means of the Boltzmann-Matano method, was fitted and the parameters allowed to reconstruct any concentration profile obtained in the same conditions. The proposed model is very effective in reproducing the ion exchange process of a single ion and, moreover, outlines several interesting questions on the correlation between the ion exchange thermodynamics and the cation site microstructure. Nevertheless, it lacks fundamental considerations about the activation energy affecting the self-diffusion coefficients.

Summarizing, ion exchange is quite a complex process, involving structural rearrangements of both ion sites and of the glass network. In this framework, even if in most cases the resulting structure does not really reproduce the structure of mixed-alkali systems, the models developed for describing MAE can be useful for modeling the inter-diffusion mechanism. In order to avoid the multiplication of adjustable parameters, a more complete analysis of the microscopic and thermodynamic aspects of the glass network changes has to be carried on. First of all, the problem has to be faced for well-established glass compositions: in fact, experimental results underline the role in the ion migration of the amount of NBOs, which depends also on the concentration of Al_2O_3 or B_2O_3 . Then, a deeper study about the interaction energies, taking into account how they are related to the formation of bonds, of local rearrangements and of NBOs, is needed. Finally, different analysis techniques can be used to evidence the changes of the glass microstructure and of the cation sites.

3. – Ion exchange in glass for optoelectronics

3.1. Ion-exchanged optical waveguides. – Ion exchange in glass is a widely applied technology for the fabrication of passive and active optical waveguides. By the substitution of Na^+ or K^+ in silicate or phosphate glasses with other ions of different size and polarizability, a refractive index change in selected surface regions can be achieved, giving rise to the formation of waveguiding patterns on planar substrates.

The production of opto-electronic devices by means of the ion exchange process is a quite established technique, which is extensively applied in industrial processes. Several reviews [93-96] on ion-exchanged waveguide technology and more than thousands of research papers connected to it have been published. Ion exchange is an alternative procedure to step-index systems obtained by thin-film deposition technologies, owing to the ease of fabrication, —it does not require a high-vacuum deposition apparatus— to the absence of interfaces, which are the cause of optical losses, and to the possibility of shaping the refractive index of buried waveguide in order to optimize the matching with optical fibers. On the other hand, thin-film waveguides, from the beginning of their appearance [97], are competitive in many applications, like multimode interference couplers, where a high-index contrast is needed, and in optical chemical sensors, where thin film of sensitive compounds are requested. Laser annealing procedures demonstrated to be suitable for the reduction of the scattering losses [98] and, moreover, thin-film waveguides are the best candidates for integrated systems on silicon devices [99] or flexible systems [100].

Ion implantation is also a tool for the production of optical waveguides and it will be described in detail in sect. 7.

The first work on ion-exchanged waveguides appeared in 1972 by Izawa and Nakagome [41]. The waveguides were obtained by ion exchange from a molten salt bath of Tl^+ with Na^+ and K^+ ions in glass. Key processing techniques were also presented: field-assisted ion exchange to achieve deeper concentration profiles and a second step introducing original ions back into the glass to bury the waveguide under the surface, achieving a low loss device. In addition to planar waveguides, multimode channel waveguides were fabricated using lithographically a patterned mask film.

In the following years Giallorenzi *et al.* [101] fabricated waveguides with purely thermal diffusion of Tl^+ , Ag^+ and K^+ , evidencing that silver produces a surface refractive index increase by one order of magnitude ($\Delta n = 0.08$) higher with respect to potassium ($\Delta n = 0.008$).

In 1974, Saruwatari and Izawa produced the first non-fiber glass waveguide laser by means of Tl^+ exchange process in Nd-doped borosilicate glass. A 4 mm long multimode waveguide capable of pulsed laser operation was achieved [102].

After these first very promising works the research on passive and active optical devices obtained by ion exchange in glass proceeded until present, exploiting different methods for the production of low loss systems suitable for the coupling to optical fiber networks. In all the devices the suitability of the ion exchange procedure is based on the control of the refractive index which increases in correspondence to the ion diffusion profile.

3'2. Relationship between concentration profile and refractive index. – In ion-exchanged waveguides the ion concentration distribution $C(x, y)$ is converted into a refractive index profile $n(x, y, \lambda)$, where λ is the light wavelength, mainly by three mechanisms: ionic size, ionic polarizability, and photoelastic effects due to induced stress [103]. The latter mechanism dominates the index increase for K^+ - Na^+ or Cs^+ - Na^+ ion-exchanged systems, while in Ag^+ - Na^+ and Tl^+ - Na^+ ion-exchanged waveguides the index increase is due mainly to the high difference in polarizability of the two ions [104, 105]. For small concentrations of the exchanging ion, there is a linear relationship between exchanged ion concentration and index change [106]. This follows from the Lorentz-Lorenz equation, which relates the refractive index n of a dielectric to the polarizability

$$(12) \quad \frac{n^2 - 1}{n^2 + 2} = \frac{1}{3\epsilon_0} \sum_i N_i \alpha_i,$$

where N_i is the concentration and α_i is the polarizability of the element i , respectively. By assuming that the exchange process substitutes an amount ΔN of element A with an identical amount of element B , the difference Δn between the index n after the exchange and the substrate index n_{sub} is

$$(13) \quad \Delta n = \frac{n_{\text{sub}}^2 + 2}{6n_{\text{sub}}} (\alpha_B - \alpha_A) \Delta N.$$

For index increases of the order of 10^{-2} the index profile can be expressed by the relationship

$$(14) \quad n(x, y, \lambda) = n_{\text{sub}}(\lambda) + \Delta n(x, y) = n_{\text{sub}}(\lambda) + \Delta n_0(\lambda) X(x, y),$$

where $X(x, y)$ is the normalized concentration profile, $n_{\text{sub}}(\lambda)$ is the substrate index, Δn_0 is the maximum index increase for a given glass/salt melt combination and coordinates x and y account for bi-dimensional diffusion profiles realized through masks in order to obtain 2D shaped profiles for the coupling to optical fibers.

Since the ion exchange process occurs at temperatures well below the stress relaxation temperature of the glass, substitution of Na^+ with larger alkali ions like K^+ gives rise to a planar compressive stress state within the exchanged layer. In fact, volume changes can be accommodated only along the direction normal to the substrate, since the surface is prevented from expanding (or contracting) laterally due to resistance to bending of the glass substrate. The accommodation results into a swelling of the surface, for the K^+ - Na^+ exchange, and causes a residual compressive stress in the exchanged layer which is balanced by a tensile stress in the substrate. Planar stress induces a planar birefringence, resulting in a difference in refractive index between Transverse Electric (TE) and Transverse Magnetic (TM) propagation modes, according to the relationships

$$(15) \quad n_{\text{TM}} = n_0 + 2c_2\sigma, \quad n_{\text{TE}} = n_0 + (c_1 + c_2)\sigma,$$

where n_0 is the pure compositional (stress free) refractive index related to the difference of ionic polarizability of the exchanging ions, σ is the stress value and c_1 and c_2 are the stress-optical coefficients.

A more detailed discussion about the stress dependence on the ion concentration will be developed in the section describing the ion exchange in glass for chemical strengthening.

3'3. Waveguide fabrication processes. – The ion exchange processes for waveguide fabrication can, in most cases, be represented as a sequence of one and three process steps. The first step is the thermal or field-assisted ion exchange from a molten salt source through a patterned mask, in case followed by thermal annealing or by a second exchange treatment for the shaping or the burying of the index profile. As previously outlined, the field-assisted procedure can be realized with different configurations: from a molten salt at the anode side; from a metal thin film deposited on the top of a patterned mask or directly patterned onto the glass surface, by means of a combination of a patterned metal film with molten salt source at the anode surface. The latter procedure can be used as a one-step process to produce buried waveguide since, after the metal film source consumption it proceeds as a field-assisted burial.

For the fabrication of channel waveguides, different mask materials with lithographical patterning have been used. The most widely used materials are aluminum [107-109] and titanium [108] but often dielectric masks are preferable to metal films, since Ag nanoparticles can form under the mask edges [107]. Moreover, conducting masks can generate in the glass an electric field affecting ion concentration distributions [107, 109]. Furthermore, in field-assisted processes a thin depletion layer forms under metal masks [110]. For these reasons, dielectric mask materials are often preferred, like aluminum oxide [107], silicon dioxide and silicon nitride [111].

Thallium ion exchange was used in the first publication in the field, and it was also used for the first industrial fabrication processes. The industrial Tl^+ - K^+ exchange procedure was developed for waveguide fabrication by Corning and first applied to multimode waveguides [112], but then product development concentrated on single-mode waveguide component products, mainly 1-to- N splitters [113]. Nippon Sheet Glass developed an alternative fully thermal process for more shallowly buried waveguides [114]. The

advantage of thallium is that a rather large index increase can be obtained, but the main reason for the limited use of this species, particularly in research work, is its high toxicity.

After the pionering works, the two most widely used ions in ion exchange glass waveguide production have been Ag^+ and K^+ . Potassium ion exchange can be used to produce low-loss surface waveguides and its low refractive index increase allows relaxed mask opening dimensions. However, in K^+ -exchanged glass compressive stress states, which largely contribute to the index increase [115], give rise to the formation of birefringent waveguides [74, 105]. Moreover, K^+ exchange is not well suited for buried waveguide fabrication, due to the low index contrast and the large difference in mobility of K^+ and Na^+ ions.

Silver ion is the most suitable species for waveguide fabrication, which explains its wide popularity. It allows to achieve a relatively high index increase, which can be controlled by melt dilution, and the mobility ratio of Ag^+ and Na^+ ions is not so low in most glasses. The main disadvantage in silver-exchanged waveguides is related to the tendency of silver ions, for high concentrations, to reduce and to subsequently aggregate into metallic clusters, giving rise to the formation of important waveguide loss effects. Many approaches have been attempted to avoid the reduction of silver ions decrease, including the use of diluted silver concentrations and of the substitution of metallic masks films with dielectric masks in fabrication processes.

From this particular point of view, glass composition plays a key role in the reduction of silver to the metallic form. For instance, particular reducing impurity elements, like multivalent Fe, As, and Sb, should be avoided. So, a careful selection led to the use of high-quality glasses, such as Corning 0211, with reduced silver particle formation.

Moreover, it has been verified that, besides the absence of reducing impurities, an optimal glass for silver ion exchange needs a very special composition [116]. In fact, in conventional silicate sodium-containing glasses, silver ions can be reduced by extracting an electron from non-bridging oxygens to which sodium ions are associated. By adding aluminum oxide into glass, the number of non-bridging oxygen sites decreases and with a molar concentration ratio $\text{Al}_2\text{O}_3/\text{Na}_2\text{O} = 1$ a glass network free from non-bridging oxygens can be obtained, and Ag^+ ions exchanged into such aluminosilicate glasses are quite stable. Boron can have a similar role as aluminum in the glass structure. So, a special aluminoborosilicate glass (BGG315,62) was developed for silver ion exchange, with a fraction of oxygen replaced by fluorine in order to reach a refractive index value similar to that of silica for the matching with optical fibers.

It has to be underlined that the main difference between small-volume laboratory research on ion-exchanged waveguides and industrial-scale production is the availability of special glass substrates developed for this purpose and, to some extent, equipment fabrication suitable for industrial quality manufacturing.

The latter is particularly relevant with respect to field-assisted processes with molten salts, which are complicated to operate conveniently and with controllability to achieve the required reproducibility. To date, no commercial manufacturing has utilized dry silver ion exchange.

At present, ion exchange allows to fabricate any passive waveguide device, including Mach-Zehnder interferometers [117] multimode interference (MMI) couplers [118, 119] and phased-array multiplexers [120].

Optical gratings within ion-exchanged waveguide structures can be obtained by means of photowriting techniques on UV photosensitive glasses like BGG3I glass [121].

The ultimate goal of integrated optics, that is the integration of multiple devices on a single substrate, is still under study. The integration of a Mach-Zehnder filter device with a 1-to-8 splitter was demonstrated by Zhang *et al.* [122] and multiple waveguide layers have been obtained by combining silver-sodium ion-exchanged surface waveguides with buried waveguides 22 μm deep [123].

At present, ion-exchanged waveguides exhibiting optical nonlinearity have been obtained by applying the ion exchange process to semiconductor doped glasses. Commercial filters with CdSe crystals were treated by Gabel *et al.* [124] giving four-wave mixing effects and picosecond all-optical switching was demonstrated by Finlayson *et al.* [125]. More recently, $\text{Ag}^+\text{-Na}^+$ and $\text{K}^+\text{-Na}^+$ ion exchange on PbS doped glasses allowed to obtain highly nonlinear waveguides [126]. The main problem in this approach is the low optical quality of commercial doped glasses or the optical losses induced by the presence of the quantum dots. As an alternative, niobium- and titanium-containing glasses gave rise to high nonlinearity after $\text{Ti}^+\text{-K}^+$ ion exchange.

Another important application of ion-exchanged waveguides is the production of rare-earth-doped waveguide lasers. In all the cases silver or potassium ion exchange has been performed on doped bulk glasses. The first device was demonstrated by Saruwatari *et al.* in 1973 [102] with a field-assisted burial process on a Nd-doped borosilicate glass. But the explosion of the interest in this kind of devices started in the early 1990s, when Nd-doped waveguide lasers and amplifiers at 1.06 μm wavelength were produced [127, 128]. Moreover, due to the emerging applications of Er-doped amplifiers or multiplexer for networks operating at 1.55 μm , the research focused on Er-doped planar ion-exchanged waveguides. In particular, the best results have been achieved using phosphate glass substrates, where rare earth can be dispersed in higher amounts, with respect to silicate glasses, without clustering phenomena which lead to reduction of conversion efficiency. The first important result was obtained by Barbier *et al.* [129], who produced an amplifier with 7 dB of net gain in a double pass configuration with a device length of only 41 mm in an Er/Yb-codoped phosphate glass substrate. The waveguide needed a field-assisted process in order to reduce the fiber-waveguide coupling loss and the propagation losses at the waveguide surface. In fact, phosphate glasses have poor chemical durability when exposed to molten salts, resulting in a poor surface quality of the device. For this purpose, special phosphate glasses with improved chemical durability were used in the following works [130]. Alternatively, Ag thin-film ion exchange process demonstrated to be feasible for the fabrication of high quality surface waveguides with high gain per length unit [131].

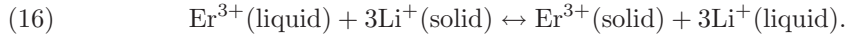
Beside optical communications, the most important application field of optical waveguides is for optical sensors. In this framework, chemical sensors are based on the interaction of the evanescent field with a thin liquid layer on the waveguide surface. Ion-exchanged waveguides exhibit low refractive index difference, so that the modes extend more into the substrate than above the surface. Evanescent field enhancement was achieved by a high index thin film on the surface, creating a dielectric loaded waveguide [132]. Alternatively, surface plasmon sensors have been obtained by means of metal films, exploiting the high portion of the evanescent field of surface plasmons [133].

4. – Ion exchange in ferroelectric materials

4.1. Introduction. – The exploding demand for internet access, telecommunications and broadband service has pushed to a higher light wave transmission capacity, forbidden to electronic technology. As a consequence, an increasing impulse towards the development of new materials for generating, guiding and processing light signals compatible

with a planar geometry is constantly given. In the last decade trivalent erbium ion has attracted great interest in integrated optoelectronics technology [134-138] since traveling wave amplifiers [139] and waveguide lasers [140] have been realized. Among different hosts, LiNbO_3 has drawn particular attention, owing to its good intrinsic electro-optic behavior and to its excellent nonlinear optical properties. Several ways have been investigated with the aim to control erbium distribution and concentration in the host crystal, as well as its site location. Bulk doping of LiNbO_3 during crystal growth has been extensively studied [141, 142] as well as local doping by erbium diffusion from thin film and erbium implantation [134, 143, 144]. More recently, it has been demonstrated that ion exchange is a suitable method for the erbium local doping of LiNbO_3 crystals [145, 146] with the possibility of tailoring optical performances of the material through preparation parameters. Ion exchange is a well-established process, used to create optical waveguides in lithium niobate by means of proton-lithium ion exchange (see [147] and references therein), and by the exchange of mono- and divalent transition metal ions (see [148] and references therein). In the present review the most important aspects of the ion exchange process will be given and it will be shown how this process can be exploited for the local doping of LiNbO_3 single crystals.

4.2. Theory of ion exchange process in LiNbO_3 . – The ion exchange process consists of a transfer of different species between phases brought into contact, as discussed in the paragraph 2.1. The chemical potential gradient drives the species exchange between the two phases in order to maintain their charge neutrality. In this case the general process regarding the ion exchange of lithium (valence state $Z_{\text{Li}} = 1$) belonging to solid phase (LiNbO_3 crystal) and erbium (valence state $Z_{\text{Er}} = 3$) belonging to a liquid phase will be considered. As a consequence, the modification of the solid phase composition and structure is generated in a surface nearby region. The ion exchange process at the interface can be described in terms of a chemical reaction



Actually, a comprehensive phenomenological model of the ion exchange process in LiNbO_3 was not completely understood in the case of not-isovalent exchange, due to the complexity of structural rearrangements during the inter-diffusion and therefore the Nernst-Planck model was normally used for describing the process. According to this phenomenological model, the ionic flux J is given by

$$(17) \quad J_{\text{Er}} = -D_{\text{Er}} \left(\frac{\partial C_{\text{Er}}}{\partial x} + \frac{C_{\text{Er}} F Z_{\text{Er}}}{RT} \frac{\partial V}{\partial x} \right),$$

where the x -axis is normal to the surface and it is directed towards the solid solution. C_{Er} represents the erbium concentration, F is the Faraday constant, R is the Boltzmann constant, and D_{Er} represents the erbium diffusion coefficient dependent on temperature and the diffusion time [128, 129]. A similar expression holds for lithium ions. From the conditions of electrical neutrality it follows that

$$(18) \quad zC_A + C_B = \text{constant},$$

where z indicates the erbium-lithium valence ratio ($z = 3$). Therefore, it is possible to correlate the concentration with the time and the thickness involved in the exchange

process [149]

$$(19) \quad \frac{\partial C_{\text{Er}}}{\partial t} = \frac{\partial}{\partial x} \left(\hat{D}_{\text{Er-Li}} \frac{\partial C_{\text{Er}}}{\partial x} \right),$$

where the interdiffusion coefficient $\hat{D}_{\text{Er-Li}}$ is expressed by

$$(20) \quad \hat{D}_{\text{Er-Li}} = \frac{D_{\text{Er}} D_{\text{Li}} (z^2 C_{\text{Er}} + C_{\text{Li}})}{z^2 D_{\text{Er}} C_{\text{Er}} + D_{\text{Li}} C_{\text{Li}}},$$

By solving eq. (19) with numeric procedures, it is possible to determine the in-depth profile of the exchanged species which depends on the erbium/lithium diffusion coefficient ratio $M = D_{\text{Er}}/D_{\text{Li}}$. Since the Nernst-Planck model did not take into account the presence of strain induced by the ion exchange occurring inside the matrix, the numerical solutions cannot describe properly the experimental results. For this reason, we introduced [150] in eq. (17) a stress contribution as follows:

$$(21) \quad J_{\text{Er, stress}} = \frac{D_{\text{Er}} C_{\text{Er}} V_{T_{\text{IE}}}}{RT} \frac{\partial \sigma}{\partial x},$$

where σ represents the stress and $V_{T_{\text{IE}}}$ indicates the volume at ion exchange temperature. A similar expression holds for lithium ion. The numerical solution of eq. (19) now depends on: 1) stress gradient profile; 2) erbium/lithium diffusion coefficient ratio, $M = D_{\text{Er}}/D_{\text{Li}}$; 3) volume ratio at ion exchange temperature (T_{IE}) with respect to room temperature (T_{RT}), $\eta = V_{T_{\text{IE}}}/V_{T_{\text{RT}}}$; 4) process temperature, T . Many trial functions, starting from a modification of those proposed for isotropic medium [149], have been used for the stress profile and finally we have observed that the most likely function should have the following gradient [151]:

$$(22) \quad \frac{\partial \sigma}{\partial x} = \frac{\partial C_{\text{Er}}}{\partial x} \left[\frac{(\eta - 1)c_{ij}}{3(1 - \nu)} \right] C_0 \frac{z^2 C_{\text{Er}} M (\eta + 1) - z(z + 1) C_{\text{Er}} + C_0 (1 - z)}{(C_0 + C_{\text{Er}} (1 - z)^2) + (C_{\text{Er}} z (zM - 1) + C_0) C_0} \cdot \frac{z C_{\text{Er}} (zM - 1) + c_0}{z C_{\text{Er}} (z - \eta) + C_0 (\eta - z)},$$

where C_0 is the lithium bulk concentration, β represents a constant depending on the effective elastic stiffness, c_{ij} , and the relative variation volume during the ion exchange process and ν is the Poisson ratio ($\nu = 0.23$ for X -cut crystals, $\nu = 0.43$ for Z cut).

In particular, $\eta = V_{T_{\text{IE}}}/V_{T_{\text{RT}}}$ can be derived from the thermal expansion theory with lithium niobate thermal expansion coefficient reported in literature [150].

The peaked shape of the stress gradient (see fig. 3, where stress gradient is reported for different processing times) limits erbium in the diffusion inside the substrate and takes into account the crystal modification induced by lithium replacement with an ion of different valence state and ionic radius. Therefore, if suitable values of M and c_{ij} are chosen, the shape of the numerical solution of eq. (19) in which the stress contribution has been added successfully fits the experimental erbium profile.

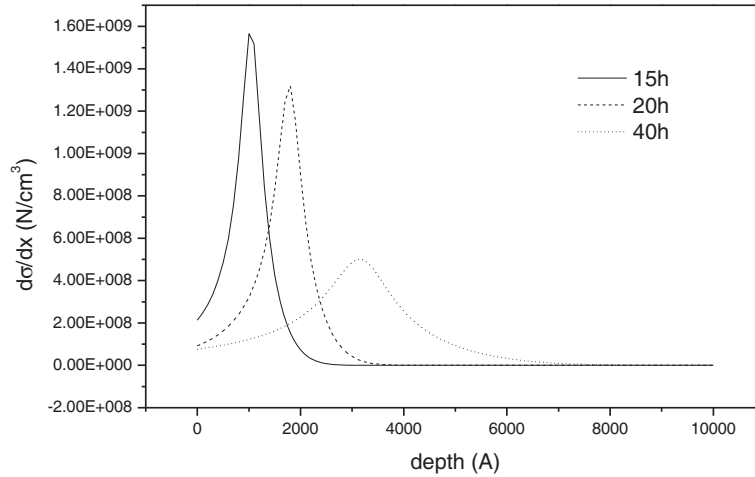


Fig. 3. – Stress gradient profile for processing times ranging from 15 h to 40 h for a X-cut crystal. Ion exchange conditions: temperature = 650 °C, $M = 6.5 \times 10^{-5}$, $c_{ij} = 2.0 \times 10^{11}$ N/m².

4.3. Experimental. – Ion exchange was performed by immersing *X*- and *Z*-cut crystal plates in a molten salt bath at temperatures ranging from 560 °C to 700 °C and processing times varying from 1 h to 40 h. The starting bath composition was $K_2SO_4 : Na_2SO_4 : Li_2SO_4$ (19.4:9.9:70.7 wt.%), to which $Er_2(SO_4)_3$ was added in various percentages, varying from 0.002 wt.% to 5 wt.%. The heating rate of the mixture of salts is slow (100 °C/h) for preventing any chemical interaction with lithium niobate surface. Secondary ion mass spectrometry (SIMS) measurements have been performed using a ion microscope (Cameca ims 4f) equipped with a normal incidence electron gun used to compensate the surface charge build-up while profiling insulating samples such as $LiNbO_3$ crystals. The spectroscopic properties of the ion-exchanged crystals have been analysed by photoluminescence in the 250–850 nm range, by using a spectrofluorimeter, equipped with a 150 W-Xe lamp. X-ray diffraction (XRD) measurements have been made by a powder diffractometer with Ni-filtered $Cu K_\alpha$ radiation for phase investigation and by high-resolution diffractometer with $Cu K_\alpha$ radiation for strain measurements.

A systematic study has been performed in order to evaluate the role of the most important parameters that could influence the ion exchange process, *i.e.*, temperature, processing time and crystal orientation.

4.4. Results. – From the investigation [145, 146], it is shown that erbium content inside the mixture salts plays a critical role in the process. In particular, neither erbium sulphate concentration lower than 0.002 wt.% or higher than 5 wt.% is effective for the ion exchange process even for long processing times (up to 200 h) and high temperatures (up to 700 °C). Moreover, for $Er_2(SO_4)_3$ concentration from 0.6 wt.% to 5 wt.%, the exchange depth is reduced by a factor two and the ion exchange appears to be anisotropic, the thickness d involved in the process being greater in *X*-cut crystals than in *Z*-cut ones ($d_X \approx 1.7d_Z$). For all the exchanged samples, the erbium experimental profile determined by SIMS analysis has been compared with the theoretical one obtained from the numerical solution of eq. (19) in which the stress contribution has been added. In fig. 4 experimental and theoretical profiles have been reported for an *X*-cut crystal (exchange parameters: $T = 650$ °C, $Er_2(SO_4)_3 = 0.2$ wt.%, $t = 40$ h).

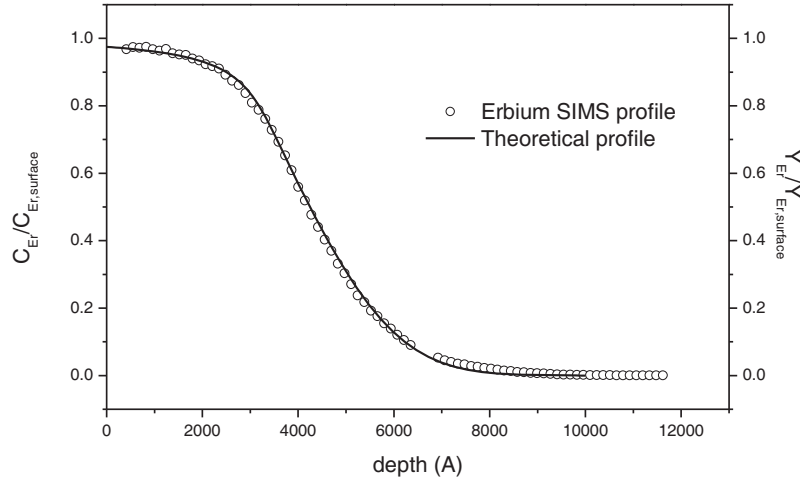


Fig. 4. – Comparison between experimental and theoretical in-depth profiles. SIMS erbium yield has been normalized to its maximum value. Ion exchange conditions: temperature 650 °C, processing time 40 h, erbium sulfate concentration 0.2 wt.%. Fit parameters: $M = 1.8 \times 10^{-4}$, $c_{ij} = 2.97 \times 10^{11} \text{ N/m}^2$.

In particular, erbium SIMS profile and the corresponding theoretical fit have been normalized with respect to the maximum erbium yield ($Y_{\text{Er,surface}}$) and concentration ($C_{\text{Er,surface}}$) respectively. By fitting the experimental data with the theoretical ones, it was observed that erbium diffusion coefficient relative to the lithium one, M , increases by a factor 3 when the processing time varies from 10 h to 40 h both for X - and for Z -cut crystal. Moreover, in the X -cut crystals M is 3 times greater than in the Z -cut ones even for increasing processing times. Furthermore, the interface between the ion-exchanged region and the substrate was fitted with a generalized Gaussian function. It was observed that for a processing time greater than 30 h the best value of α is $\alpha = 3.9 \pm 0.9$, correlation factor 0.998 for X -cut and $\alpha = 3.4 \pm 0.5$, correlation factor 0.999 for Z -cut. Therefore, it emerges that in the ion exchange process the interface is sharper than that obtained by erbium thermal diffusion from a thin film deposited on the surface, since in the last case the found value α is 2 [146]. The confidence level of the model proposed in [151] was confirmed not only by the comparison with Er^{3+} ion exchange but also with other not-isovalent exchange (such as in chromium and other rare earth incorporations) suggesting that the introduction of the role of the stress in the process plays a key role.

From the structural point of view, XRD analyses have revealed no other phases than LiNbO_3 and weak strains only (lattice mismatch less than $6 \cdot 10^{-4}$ in the direction normal to the surface). It is worth noting that both the shape and position of the XRD peaks on rocking curves are independent of the exchange time, thus suggesting that the structure modification inside the LiNbO_3 crystals is the same.

Spectroscopic measurements have been performed in order to understand how the optical properties of erbium are affected by concentration quenching. Room temperature PL spectra of all ion-exchanged samples have been measured at various excitation wavelengths. The lines of the spectra are due to transitions between Er^{3+} state manifolds, corresponding to crystal field splitting into the $(J + 1/2)$ Stark levels of the free ion $4f^{11}$ electronic configuration [152, 153]. A direct identification of the states can be obtained from the results reported in ref. [153]. In particular, fig. 5 presents the luminescence

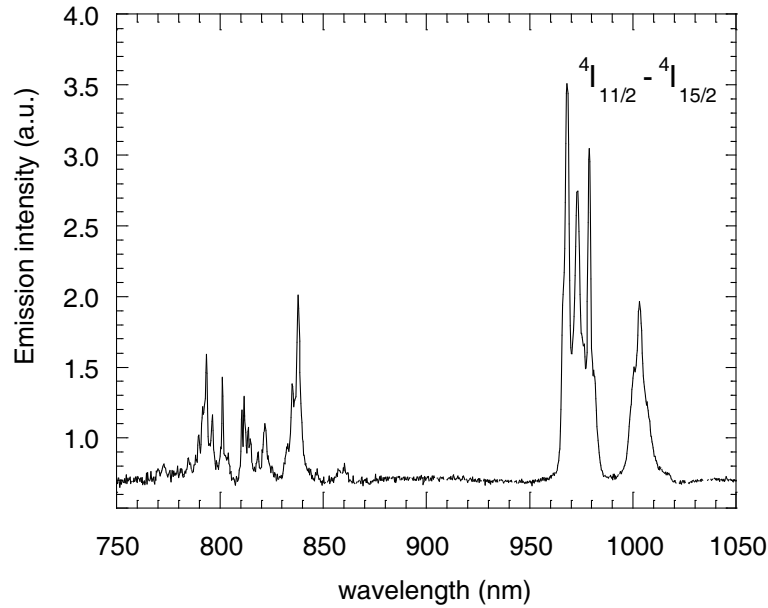


Fig. 5. – Emission spectrum of an Er : LiNbO₃ X-cut sample after excitation at 632.8 nm. Ion exchange conditions: temperature = 650 °C, processing time = 40 h, erbium sulfate concentration = 0.2 wt.%).

intensity of the ${}^4I_{11/2} \rightarrow {}^4I_{15/2}$ transition induced by erbium excitation at 632.8 nm (ion exchange parameters: temperature=650 °C, time=40 h, erbium sulphate concentration=0.2 wt.%). A linear dependence of luminescence intensity on erbium concentration was observed for erbium sulphate content below 0.6 wt%, suggesting that, in this range, concentration quenching is not appreciable.

5. – Quantum confinement in Ag exchange in glasses

Ion exchange process has been used in order to create Ag nanoclusters in Er-Yb activated glass, one of the most common systems used in integrated optics [154]. Such research was connected to the importance in the field of optics of metal nanoparticles embedded in rare-earth activated glasses, a material of interest for telecommunication application. Indeed it was shown that the photoluminescence (PL) intensity of optically active erbium ions positioned in close proximity to anisotropic Ag nanoparticles is significantly enhanced if the nanoparticles support plasmon modes that are resonant with erbium emission [155,156].

Moreover the erbium fluorescence sensitization may be also induced by ultra-small metal nanoclusters incorporated in Er-doped silica, not correlated to plasmonic process. Recent experimental works [157-162] on the occurrence of enhanced erbium PL induced by Au or Ag aggregates, produced by low fluence implantation, showed that the sensitization process was connected to an energy transfer process (ET) critically depending on the evolution of the early stages of metal clustering process starting from dispersed metal atoms up to the formation of nanometric clusters. An important aspect of ref. [154] work was to understand the effect of different Ag concentrations, introduced by ion exchange

process, on the electronic structure of silver nanoclusters in a glassy network. X-ray Photoelectron Spectroscopy (XPS) and Transmission Electron Microscopy (TEM) measurements were performed in order to investigate quantum confinement of electrons. Ion exchange process was obtained by keeping the glass at 390 °C for 67 h in a molten salt bath of NaNO₃, containing a variable molar concentration (0.5%, 1% and 5%) of AgNO₃. The formation of Ag nanoclusters has been evidenced for 5% Ag molar concentration by the presence of an absorption peak at 413 nm arising from surface plasmon resonance [5].

TEM analysis evidenced a large quantity of nanoparticles with an average diameter of 2 nm and a size distribution between 1 to 5 nm. For lower molar AgNO₃ concentration the predominance of oxidized silver with respect to silver nanoclusters determines the absence of a plasmon band in the absorption spectrum. Higher silver concentration causes Ag condensation into the nanoparticles and the chemical oxidation effects become marginal. A good agreement was also obtained by comparing the XPS analysis with TEM results. For low Ag concentrations the Auger parameter, α , is very close to the value reported in the literature for silver oxide. A slightly large value of α falls between those of silver oxide and pure bulk silver. These results confirm that XPS spectral changes correspond to well-defined chemical and structural organization of silver in soda-lime glasses. Moreover these results are in agreement with the estimates obtained from the plasmon resonance frequency and TEM analysis.

6. – Ion and laser irradiation of ion exchange glasses

Glasses containing transition metal clusters attract attention from many scientific aspects: research in the cluster properties and in possible applications of such clusters for magnetic, nonlinear optics gas sensors and optoelectronics purposes [6, 7]. Nanometer clusters of noble metals, as Au and Ag, exhibit in glasses a strong absorption in the visible spectral range (SPR) [5, 163]. The possible applications of such metallic nanoclusters have led to an interest in investigating the binary exchange technique, as discussed in sect. 2, for introducing the respective metals into the glass, already used since centuries in making colorful glasses. The process presents various steps: 1) introduction of metal atoms into the glass by ion exchange process, which results in a non-metallic chemical environment; 2) transformation in metal; 3) cluster formation with control of size, shape and arrangement. The subsequent treatments in order to reduce the elements incorporated in the glass in oxidic environment to metallic state are annealing in reducing atmosphere, laser or ion irradiation. Irradiation with swift heavy ions showed a “directionality” of metallic cluster formation: metallic silver clusters were observed in a chain-like arrangement along the direction of the ion beam [164, 165]. Such result has been attributed to the higher energy density transferred to the electronic system of the material, above the threshold for ion track formation in glasses in the case of irradiation with ions Au and Ag. Ion irradiation of metal-doped glasses for the controlled promotion of metal particle growing has drawn considerable attention in the last years [166, 167]. The glass is irradiated with low-mass ions (He, for instance) or protons. The electronic component of the stopping power, which is the dominant mechanism of energy loss in such irradiations, triggers metal clusters precipitation due to the energy inelastically transferred to the system. The metal oxide can be introduced in the glass either during its thermal synthesis (as in ref. [166] in which 0.08 wt% Cu₂O oxide was present inside a soda-lime glass) or it can be obtained by an intermediate step which involves ion exchange between alkali ion inside the glass with the noble-metal ions (as in refs. [167, 168] in which Na⁺ $\leftrightarrow$ Ag⁺ ion exchange was performed in a soda-lime glass). As

regards this last approach, the technique of ion exchange is very effective in doping much thicker layers with respect to ion implantation: few microns thick optical waveguides can, for instance, be easily obtained with an increase of the linear refractive index useful to confine the light within the modified layer. The samples were obtained as follows: a soda-lime glass was preheated and immersed for 30 min in a molten salt bath of molar concentration 0.1% AgNO_3 in NaNO_3 at 320°C [168]. A close agreement was observed between silver concentration and refractive index increase. Irradiations of the exchanged samples were performed using He^+ , Li^+ and Ne^+ ions at the energy of 100 keV. The ion fluence was 2×10^{16} ions/ cm^2 , except for helium which was implanted at fluences in the range 10^{16} – 10^{17} ions/ cm^2 . For Ag^+ -exchanged soda-lime glass, after room temperature irradiation with 100 keV He^+ ions, by increasing the He^+ irradiation fluence, the SPR absorption band, centered at about 410 nm, is more and more evident: this is due to the precipitation of spherical Ag nanoclusters in the irradiated region. Rutherford Backscattering measurements indicated that no remarkable difference can be found in the silver concentration profile before and after the irradiation at room temperature: this indicates that the nanocluster nucleation occurred uniformly in the whole irradiated volume.

The role played by the different ion species is also important: in fact, in the case of lithium irradiation, the absorption band for substrate temperatures of 90 and 150°C is more intense than for the helium-irradiated specimens. Since only the collisional part of the energy deposited in the sample by the incident beam is appreciably different in the two irradiations, we can infer that the higher radiation damage caused by the lithium ions favors, in this intermediate temperature range, the migration and aggregation of silver atoms. Electron irradiation produces a silver surface depletion, accompanied by a silver accumulation towards the sample interior, at a depth roughly corresponding to maximum primary electron range. Such in-depth migration is due to the formation of an electric field, attained in an interval time of about 10^{-2} s [169]. The electric-field-assisted diffusion is characterized by a radiation enhanced diffusion coefficient, some order of magnitude greater than the thermal diffusion coefficient quoted in the literature [170]. It is interesting to underline that in the case of proton bombardment silver migration occurs from the interior towards the surface, due to the formation of an electric field in the opposite direction with respect to that induced by electron irradiation. Such behavior was observed for alkali migration in proton-irradiated soda-lime glass and explained in terms of electric-assisted diffusion [170, 171]. The samples display an optical absorption band peaked at about 410 nm, a typical absorption value in metallic silver nanoclusters, related to SPR. Similar result has been obtained by helium irradiation of silver exchange process in AR Schott glass by J. Blondeau *et al.* [172].

Excimer-laser (20 ns pulse duration and 248 nm wavelength) irradiation of silver-exchanged soda-lime glass determines the formation of Ag nanoclusters whose average size depends on the laser energy density (20 ns pulse duration and 248 nm wavelength) [173]. The Ag^+ reduction should be related to electronic processes or thermal effects both promoted by the laser irradiation with the most probable final result: $\text{Ag} + \text{NBO} \rightarrow \text{Ag}^0 + \text{NBO}$, where NBO stands for non-bridging oxygen. Moreover, because of the timescale involved in the experiment (tens-hundreds ns) Ag atoms move with diffusion coefficients values appropriate to a liquid phase, say 10^{-5} – 10^{-4} $\text{cm}^2 \text{s}^{-1}$. To attain a liquid phase the authors [174] suggested absorption mechanisms related to electronic defects produced in the first ns of the laser pulse. When both Ag^0 and a liquid phase are obtained, then the dynamics of the resolidification front, with a velocity of order of 1 – 2ms^{-1} , may help in understanding the clustering of the Ag^0 atoms. These results indicate that by combining ion exchange and ion or laser irradiation techniques,

it is possible to induce and control the formation of silver clusters, of nanometer size, within light wave-guiding structures. A comparison between nanoparticle formation in Ag-doped glass films by Ag-Na exchange of different glasses in molten salt baths, by thermal treatment in air and/or laser irradiation in the near UV region has been presented in ref. [174]. Depending on the glass composition and annealing temperatures, the thermal treatment was able to induce at annealing temperatures below 400 °C the formation of aggregates of few silver ions, as $(\text{Ag}_3)^{2+}$ trimers, with a PL around 600 nm considered as one of the intermediate species before the precipitation of metallic clusters [175]. Increasing the annealing temperature the intensity for this PL band increases up to 450 °C, then it drops down revealing the massive precipitation of metallic nanoparticles at the expense of $(\text{Ag}_3)^{2+}$ trimers. The silver precipitation starts with a mutual approach of silver atoms and then with the formation of “seeds” formed by few atoms that grow by increasing the annealing temperature. The authors used a different approach for Ag-exchanged glass processing by irradiating the metal-doped layer with a laser beam. Increasing the number of pulses ($\lambda = 355$ nm, pulse of 10 ns at 2 Hz of repetition rate, energy up to 45 mJ), ranging from 5 to 40 (nominal total energy deposited from 0.23 to 1.8 J) a progressive increase of the SPR band in the blue part of visible spectrum is observed, indicating silver precipitation to form metallic nanoparticles. Extended X-ray absorption fine structure (EXAFS) and Grazing Incidence X-ray diffraction (GIXRD) analyses have been used to determine the silver local environment in silicate glasses doped by the Ag alkali ion exchange process, followed by different treatments, as ion irradiation, laser irradiation and thermal annealing in reducing atmosphere [176]. Interatomic distances, coordination number and Debye-Waller factors resulted from EXAFS analysis. In as-exchanged samples, Ag atoms are expected to be bound to non-bridging O atoms, after the substitution for the Na atoms in the glass matrix. Nevertheless the obtained Ag-O bond length is 2.16 Å, significantly shorter than the Na-O bond length in soda-lime glass (2.32 Å) and longer than the Ag-O bond length in Ag_2O (2.05 Å). Moreover this bond length agrees with previous investigations [68, 84]. After laser irradiation (single shots from a ND:YAG laser, 10 ns long at 532 nm of wavelength and with energy density of 0.5 J/cm²) Ag coordinates both oxygen and silver atoms. The first shell is attributed to correlation with oxygens, with an Ag-O distance larger with respect to the one found for the as-exchanged sample. The second shell Ag-Ag distance and the coordination number increase. The interatomic distance value obtained (2.76 Å) suggests a situation in which most part of the reduced silver is in the form of very small aggregates and multimers [177] while part of the silver undergoes clusterization, as indicated by the presence of a detectable SPR band in the optical absorption spectrum. Helium irradiation (energy 1.5 MeV, fluence 2.10^{16} ions/cm²) determines an effective reduction and aggregation of metallic dopants inside superficial layers of glass, as suggested by the presence of SPR at 400–420 nm. The observed absorption is consistent with indications of Mie theory for silver clusters having radii of few nanometers. In the dense electron cloud created by incident radiation, Ag^+ are converted to atomic silver that diffuse and precipitate to form clusters. In the first shell Ag coordinates oxygen atoms at 2.08 Å, comparable to the Ag-O distance in Ag_2O . The second shell, Ag-Ag coordination, indicates an Ag-Ag distance (2.86 Å) comparable to the Ag bulk (2.88 Å) and a low coordination number, due partly to the cluster dimension (high surface-volume ratio). The thermal annealing at 250 °C of the exchanged sample in reducing atmosphere determines the formation of a multimer structure [178] as suggested from the second shell distance value (2.70 Å), and to an oxide phase. The precipitation of metallic atoms is interpreted on the basis of the so-called “tarnishing model” [178]. GIXRD analyses indicate that in helium ir-

radiated sample the lattice parameter of crystalline silver clusters is the same of bulk material (4.090 Å), while the diameter of particle is 83 Å. A subsequent laser irradiation on the implanted sample promotes the reduction of cluster size (60 Å) and a contraction of lattice parameter (4.078 Å).

7. – Ion beam processing

Modern photonics is based on methods of optical processing that require signal routing via optical waveguides pathways. The essential step, as previously discussed, in defining an optical waveguide is to have a higher refractive index than the surrounding material, and, alternative routes to this have been to use deposition of high index layers, diffusion doping by introducing new ions of greater polarizability, or out-diffusion of component ions. All the routes have degrees of success in limited systems, but guide formation does not automatically retain the desirable optoelectronic or nonlinear properties that existed in the original bulk material. Less obvious problems are that deposited layers may be strained or include columnar growth and diffusion doping may require a very long development phase. Ion implantation offers several alternative ways to make waveguides. The first is that it can be used to implant impurities. The advantages are that the process can be conducted at any selected temperature and by suitable masking, the choice of ion energy and dose can directly define the waveguide patterning. In a comparison with ion exchange process, ion implantation technique presents some peculiarities, but also some disadvantages. A fundamental feature of ion implantation is that the process does not take place under thermodynamic equilibrium and consequently the usual solubility limits of the implanted ions in the host can be largely overcome, achieving impurity local concentrations inaccessible by conventional synthesis routes. In particular metastable alloys, by subsequent implants of different species, may be formed with peculiar chemical-physical properties [6, 8]. Such technique showed interesting results in the modifications of mechanical properties (hardness, oxidation, wear, electrochemical and corrosion fatigue behavior) of metals, steels, but also glasses. However the use of low-energy accelerators (20–400 keV), that permit to obtain high irradiation current and consequently high implanted doses in acceptable irradiation times, important for industrial applications particularly in the field of metallurgy, limits the depth of implanted ions and consequently the thickness of the modified matrix layer. Ion exchange technique can modify a material composition at high depth, for long time treatments, but limitations are due to matrix composition, process temperature, typology of exchanged ions.

Several waveguide fabrication processes have been developed to obtain structures in LiNbO_3 crystals, as titanium in-diffusion [179], ion exchange, as previously discussed in sect. 4 and light ion implantation [180, 181]. In the past the first method was preferred due to easy feasibility and high reproducibility shown by the local doping techniques. However the possibility of introducing any given impurity at low temperature with accurate control of both the depth and lateral concentrations of the dopant makes ion implantation process an attractive technique for waveguide formation. There are two main contributions to the ion beam modification process. One of them, usually defined as irradiation damage, is associated to the interaction of the incoming ions with the material; *i.e.*, to the energy deposition. The other contribution to the modification of the material has to do with the incorporation of those bombarding ions into the structure of the material, *i.e.* ion implantation. The irradiation with beams of ions of medium mass and high energy results in the separation of the region of strong electronic damage from that associated to nuclear damage and implantation. The possibility of introducing any given impurity

at low temperature with accurate control of both the depth and lateral concentrations of the dopant makes the implantation process an extremely attractive technique for waveguide formation, in comparison with ion exchange process. A correlation between the damage profile and the refractive index is evidenced in insulating materials. In the first part of ion trajectory where electronic interaction dominates a damage component that is responsible for an increase in the refractive index is produced. On the contrary the collision cascade gives rise to damage that causes lowering of refractive index [182]. A peculiar advantage of ion implantation process, in comparison to ion exchange, lies in the possibility of obtaining spatially defined regions of high refractive index surrounded by relatively lower refractive index matrix, with low degradation of the opto-electronic and transmission properties of the base material and introducing detrimental tilt of the optical mode polarization plane. By using low dose, medium-mass ion implantation at several MeV energy in insulating materials, as LiNbO_3 , it is possible to produce very good optical waveguides and other photonic devices requiring that at least one of the dimensions scales with the light wavelength [182, 183]. The refractive index profile can be tailored as a function of the optical needs through the choice of the ion dose and bombarding energies, with very low degradation of the electro-optic coefficient. The application of high-energy ion implantation allowed to fabricate a new class of Mach-Zehnder micro-interferometers on LiNbO_3 [184]. Such device has been tested for gas trace detection obtaining a sensitivity of about 10 ppb for NO_2 detection.

In the last years metal nanoclusters (NCs) embedded in insulating materials have received increasing interest due to their peculiar optical, magnetic and catalytic properties when the size becomes comparable to or less than the electronic mean free path [6, 8, 185, 186]. Glass-based composites play an important role as materials for various nanotechnology applications, due to low cost, ease of processing, high durability, resistance and high transparency, as well the possibility of tailoring the behavior of the glass-based structures. Plasmonic nanostructures with tailored shapes and/or composition are able to produce intense local-field enhancements, which can be further increased by exploiting interaction among the plasmonic nano-building blocks. Moreover, the local-field enhancement effects that occur in the nanoparticles amplify their nonlinear optical (NLO) response and make these systems very promising for molecular sensing applications and as ultrafast nonlinear optical elements of great interest for example for nonlinear optical applications, since they can be effectively exploited to design plasmonic nanosystems with amplified nonlinear optical response [187].

Among the different techniques developed for synthesizing and processing plasmonic nanostructures in insulating matrices, ion implantation/irradiation proved to be a very versatile tool since the functional properties of the nanostructures can be properly tuned by a tailored control of the irradiation parameters. The composition of clusters can be varied easily by sequential ion implantation in the matrix of two different elements whose energy and dose can be tailored so as maximize the overlap between the implanted species and to control their local relative concentration. Moreover NCs dispersed in dielectric matrices may exhibit superparamagnetic properties, enhanced coercivity, shift of hysteresis loop and large magneto-transport properties. Nanostructured materials play an important role in catalysis, where high surface-to-volume ratios are required.

Ion exchange process in combination with laser or low-mass ion beam irradiation, as previously discussed, may give specific contributions in the field of nanotechnology, considering the possibility to create nanoparticles at high depths. Moreover a limitation in the cluster composition, size and core-shell structure, important for catalysis, is evident, considering the characteristics of the technique.

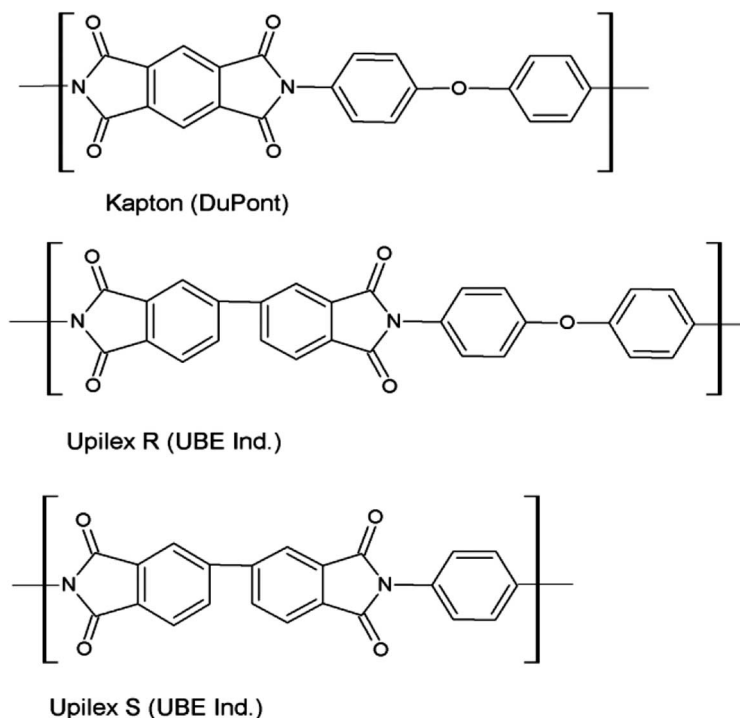


Fig. 6. – Chemical structures of some commercial polyimides.

8. – Metal-polyimide hybrids: formation of metallic nanoclusters by ion exchange

8.1. General aspects. – The entrapment of metallic nanoparticles within a polymeric matrix is a topic of great interest, owing to the possibility to obtain a composite material where the unique features of the flexible, insulating, transparent matrix are combined with the versatile optical, mechanical, electrical properties of metallic nanoparticles. In particular, within the vast landscape of polymer materials, the class of polyimides is required for applications where harsh environments, as for electrical discharges, high temperature, chemically reactive media or high vacuum, are involved.

Aromatic polyimides such as Kapton (DuPont) or Upilex (UBE Industries), whose chemical structures are reported in fig. 6, are routinely used in industrial applications [188] as dielectric materials with high breakdown voltage and a wide range of usage temperatures (from -270°C up to 400°C) without appreciable changes of tensile modulus or elongation at break. Moreover, polyimides are completely inert towards most organic solvents, concentrated acid and alkaline solutions.

However, polyimides display some remarkable drawbacks: lack of optical transparency in the UV-Vis range and of adhesion with thin metallic coatings deposited by PVD (Physical Vapour Deposition). In aromatic polyimides the former shortcoming derives from the formation of charge transfer complexes; Dine-Hart and Wright originally proposed that Charge Transfer (CT) can occur in polyimides, on the basis of colour, melting point and solubility data of model compounds [189]. After this work, many researchers de-

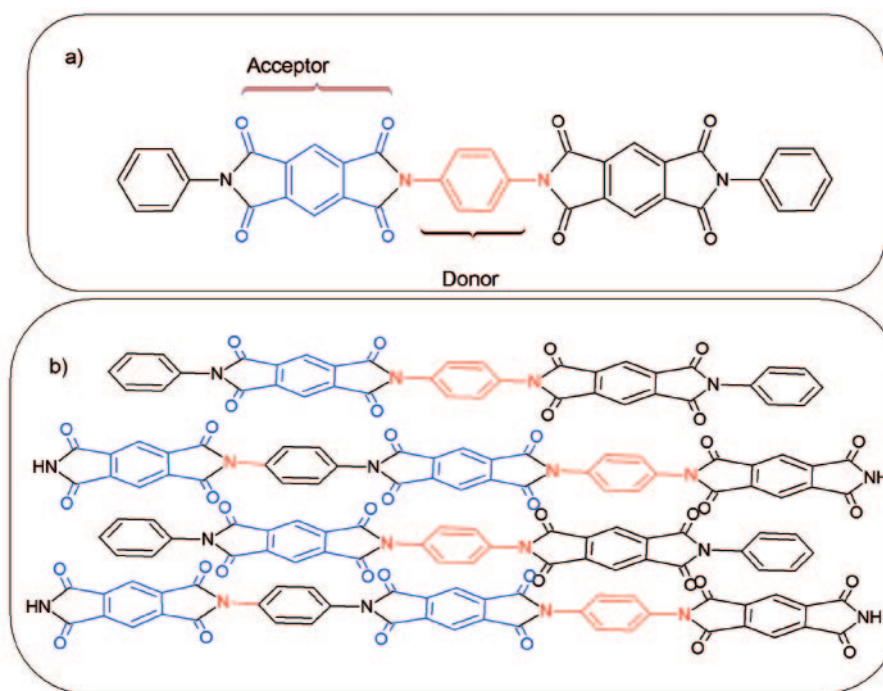


Fig. 7. – Charge transfer complexes in polyimides: intramolecular (a) and intermolecular (b) structures.

scribed aromatic polyimides as chains of alternating segments of electro-donating (amine phenyl fragment) and electron-accepting (diimide fragment) units in which either intramolecular or intermolecular CT occurs (see fig. 7). Furthermore, CT theory has been used to explain a number of properties of polyimides such as colour, photodegradation, fluorescence, glass transition and melting temperature [190].

As for optical transparency, according to Hasegawa [191], the optical properties of polyimides are strongly influenced by the weak interchain CTCs (charge-transfer complexes), which spontaneously form as two different polymeric chains closely approach. In general terms, a low ionization potential of dianiline favours the shift of the absorption edge of the final polymer toward longer wavelengths according to Mulliken's theory, thus compromising transparency. A viable strategy to increase transparency may be to hinder the formation of interchain CTCs. This can be achieved by preventing a tight stacking of the polymeric chains: monomers featuring very bulky functional groups, such as $-\text{C}(\text{CF}_3)_2-$, $-\text{SO}_2-$, $>\text{C}=\text{O}$ and $-\text{C}(\text{CF}_3)(\text{C}_5\text{H}_6)-$ such as 6FDA, 6FDAm, 3,3'-DDS, DAB and 3F can be employed (see fig. 8 for chemical structures of these monomers). These functional groups have a strong electron withdrawing capability, which contributes to transparency improvement as they are placed on dianilines (*e.g.*, in 6FDAm, DAB and 3,3'-DDS), as demonstrated by the transmission spectra obtained from different polyimides reported in fig. 9 [192].

As a matter of fact, 6FDA-bearing polyimides are almost colourless, due to the presence of hexafluoroisopropylidene $-\text{C}(\text{CF}_3)_2-$ groups. It is well known [191,193] that this functional group provides polyimides with exceptional transparency, due to the combi-

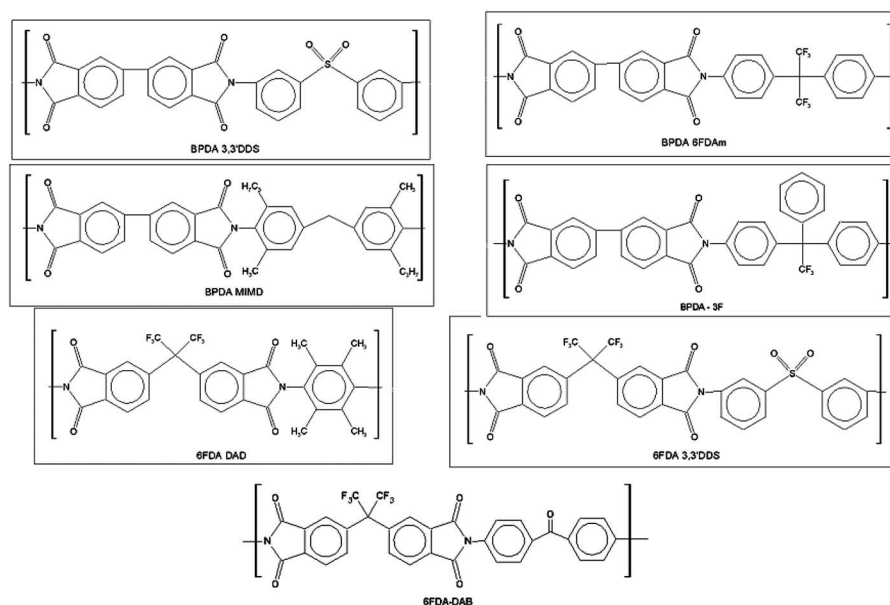


Fig. 8. – Chemical structures of polyimides with different functional groups on the diamine and dianhydride portions [192].

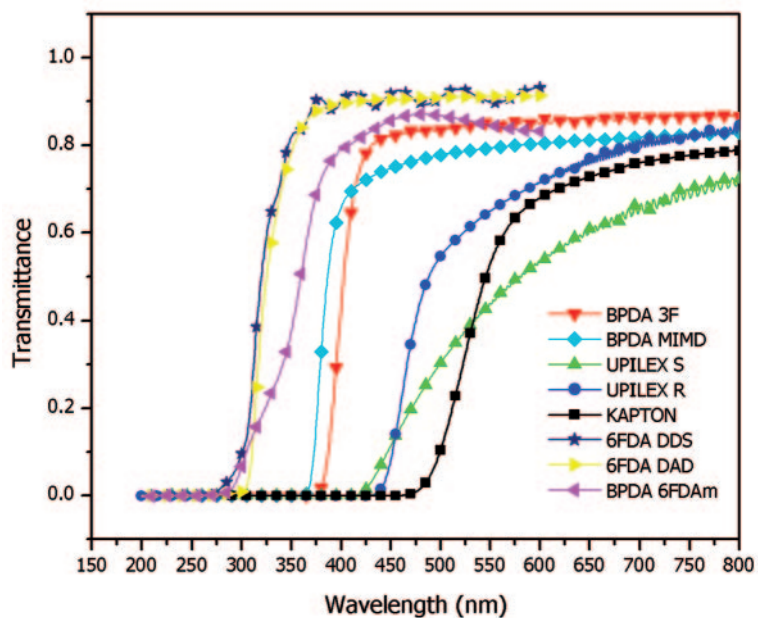


Fig. 9. – Transmittance spectra of films (thickness $25\ \mu\text{m}$) of some synthesized polyimides (see fig. 8 for chemical structures) and of some commercial polyimides (data from ref. [192]).

nation of several factors. The bulky $-\text{CF}_3$ groups act as chain-spacers along the polymeric molecules, so that they perturb the formation of intermolecular CTC. Moreover, these moieties break the conjugation between adjacent phenyl rings, thus inhibiting the electron density delocalization, which is more pronounced in the case of BPDA-derived polyimides, where the adoption of a coplanar conformation between the benzene rings is possible. The inhibition of electron density delocalization negatively affects the formation of intramolecular CTC, which are also responsible for the coloration of polyimides.

Among polyimides bearing the BPDA unit, those derived from BPDA and 6FDA are the most transparent ones, owing to the presence of hexafluoroisopropylidene groups which positively influences the transmission features, as above explained.

As for their poor adherence with PVD metallic layers, the chemical inertness of polyimides accounts for this specific feature, which is usually accompanied by a remarkable hydrophobic character, which in turn detrimentally affects the applicability of these materials in biomedical devices.

8.2. Wet chemical modification of surface properties. – Polyimides are valuable materials for both microelectronic and aerospace applications as insulating and flexible substrates for printed circuit boards, for non-metallic thin and flexible heaters and as wrapping films for semiconductor devices packaging. Unfortunately, the lack of adhesion with surface metallic films severely limited their application in these fields. Several methods were adopted in the aim to modify the surface properties of the aromatic polyimides, such as surface resistance and wettability, while preserving their outstanding bulk properties. The most effective techniques are ion implantation [194–197], gamma and electrons irradiation, glow discharge plasma treatment [198, 199] and plasma-based ion implantation [200].

Though all these methods can be successfully adopted, they are undoubtedly costly, requiring at least a vacuum plant for being carried on. Moreover, the possibility to modify a specific physical property to a defined extent is enabled by demanding experimental parameters (*i.e.* for ion implantation the implanted dose, the current, the local temperature, the ion beam energy, etc.).

Wet chemical modification of polyimides is a non-expensive technique to change the surface chemical structure of the polymer and, in turn, its physical and chemical features. The reactivity of the imide ring towards strong base, such as KOH or NaOH, is used as reported by Ing and Manske [201]. To increase the adhesion with metallic coatings, Lee and co-workers [202, 203] reported a pioneering work on the exploitation of the nucleophilic attack of strong alkali towards carboxylic moieties of polyimides, namely PMDA-ODA and BPDA-ODA. The mechanism, whose kinetics as for the specific case of PMDA-ODA has been thoroughly studied firstly by Thomas [204] and, afterwards, by Stephans and co-workers [205], is schematically shown in fig. 10a for a generic aromatic polyimide. Aiming at increasing adhesion between polyimide and PVD deposited metal layers, several authors proceed with acid treatment of the polyamate ion using mineral acids, such as hydrochloric acid, or organic acids, such as acetic acid, promoting the formation of polyamic acid, as described in fig. 10b. As far as kinetic aspects are concerned, the first part of the whole process, leading to ion-doped polyimide surfaces, is a nucleophilic attack of hydroxyl anion to the carbonyl carbon of the 5-membered imide ring. The extent of nucleophilic attack of hydroxyl anion to imide carbonyl is affected by several factors, such as temperature and time of treatment, molar concentration of NaOH or KOH solution, presence of alcoholic solvents, electron withdrawing substituents on the imide ring, permeability towards water based solutions of the pristine polyimide, which in

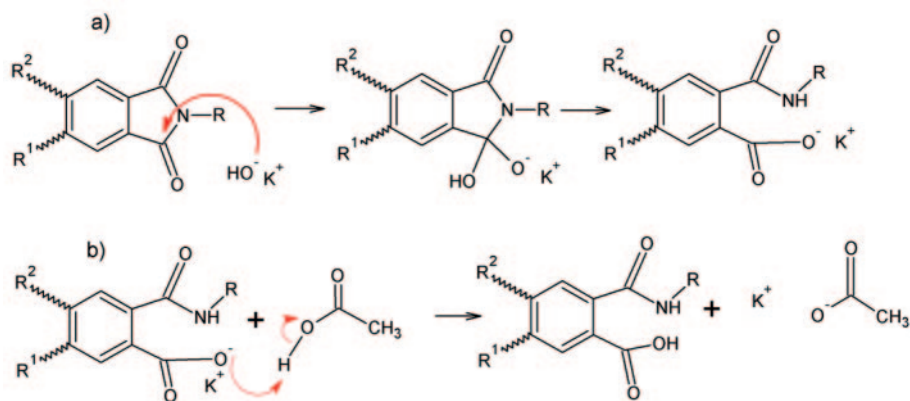


Fig. 10. – Nucleophilic attack to carbonyl moiety of the imide ring in alkaline media (a); ion exchange with acidic protons (b).

turn is affected by chemical affinity and diffusion of the nucleophile reactant through the macromolecules. Several studies focused on elucidation of the mechanism of interaction between OH^- and the polyimide network; the work of Stephans [205] can be considered as a comprehensive review of all the previously performed researches on this topic. Samples of PMDA-ODA have been exposed to hydrolysis in NaOH at different concentrations, for different times. Afterwards, surface neutralization was achieved by treatment in diluted acetic acid, thus producing the polyamic structure, depicted in fig. 10. The concentration of NaOH has been kept at quite low levels (between 0.02 and 0.17 M) by the authors, in order to avoid possible etching of the surface, which is an extreme result of treating polyimides in highly aggressive alkali solutions (high concentration and high temperature). Thus, the surface contact angle of newly produced polyamic acid surface has been measured in water solutions at different pH (contact angle titration technique), in order to quantify the conversion of polyimide to poly(amic acid). This way, it has been possible to derive the number of hydrolyzed surface sites and deduce the kinetic profile of the reaction, described in fig. 10a. Using a large excess of NaOH, pseudo-first-order kinetic data were obtained. Upon extended hydrolysis, the system reaches a limiting concentration of modified polyimide [poly(amic acid)]. Formation constants were evaluated to be quite small, and this is consistent with the existence of a self-limiting poly(amic acid) phase observed during alkaline hydrolysis under more aggressive conditions, such as those used for etching. As observed by other authors [206], upon mild reaction conditions and heterogeneous phase, which is the most frequently encountered case, a hydrolyzed and solvent swollen polyimide layer with gel-like appearance is formed indeed, whose removal is not straightforward, in spite of the ionic nature of the polyamate, and often vigorous stirring of the solution is required to bring to light the buried unmodified polyimide surface. The presence of this surface layer of polyamate accounts for the observed *diffusion limited* mechanism of wet etching, where the penetration of $[\text{OH}^-]$ reactive species through this hydrolyzed imide layer is rate determining.

The newly formed polyamic acid surface displays enhanced wettability, as proved by contact angle measurements [203, 207, 208], and improved adhesion towards both polyimide overlayers (polyimide/polyimide interface [203]) and metal films [209, 210].

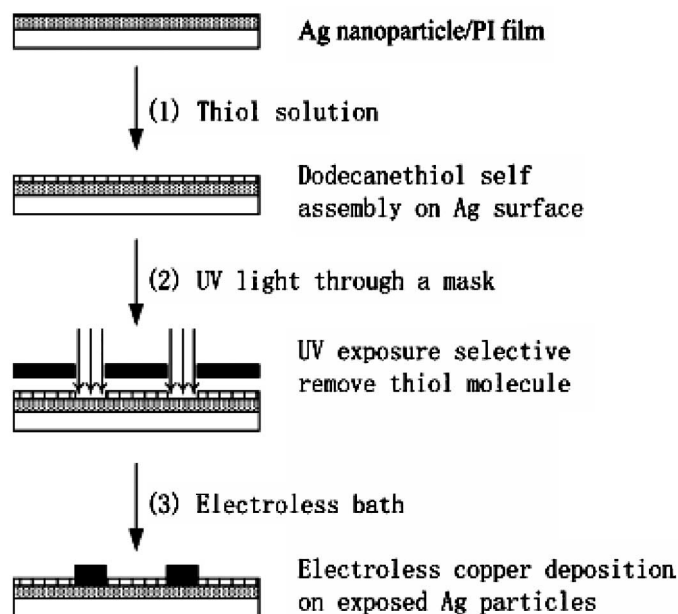


Fig. 11. – Scheme of the electroless copper plating procedure, used by Li and co-workers (reprinted with permission from ref. [209]).

8'3. Ion exchange in polyimides: electroless metal deposition, surface metallization and nanoclusters entrapment. – To enhance the adhesion between metallic coatings and polyimide substrates, Plechaty and Thomas [211] proposed to exploit the ion exchangeable sodium polyamate groups to introduce metal ions into the first monolayers of polyimide, thereby providing a hybrid interlayer as adhesion promoter. Freshly hydrolyzed polyimide in alkali is treated with acetic acid, then ion exchange with Pd^{2+} containing solution is performed. Subsequent treatments with different concentrations of Na^+ solutions, at different times and temperatures allow the ion exchange induced extraction of Pd^{2+} , thereby allowing to study the mechanism of ion exchange, which was proved to be *diffusion-controlled*. The possibility that ion exchange process is reaction-controlled was indeed ruled out from the experimental evidence that changes in the concentration of buffered Na^+ solutions did not result in changes in released Pd^{2+} ions. Moreover, from kinetic and capacity data, the depth of hydrolysis was seen not to be a linear function of time, though a certain linearity is observed at early times (pseudo-first-order kinetics), but it proceeds in a homogeneous fashion leaving a boundary between polyimide and polyamic acid. The solution to Fick's equations was used to obtain the kinetic data of diffusion. However, the diffusion coefficients obtained were found to vary with time as a result of unequal charges and diffusivities of the two diffusing species, palladium and sodium. This deviation from simple Fickian diffusion is due to the establishment of an electric field gradient in the modified polyimide during the course of ion exchange. In a general sense, the effect of an external force on diffusion has been addressed by Nernst-Planck equations in which the effect of inter-diffusion and non-equivalent charge is accounted for, as described above in subsect. 4'2.

Electroless copper plating has been successfully applied to polyimide films surface activated by tin and palladium compounds through ion exchange [212]; electroless silver

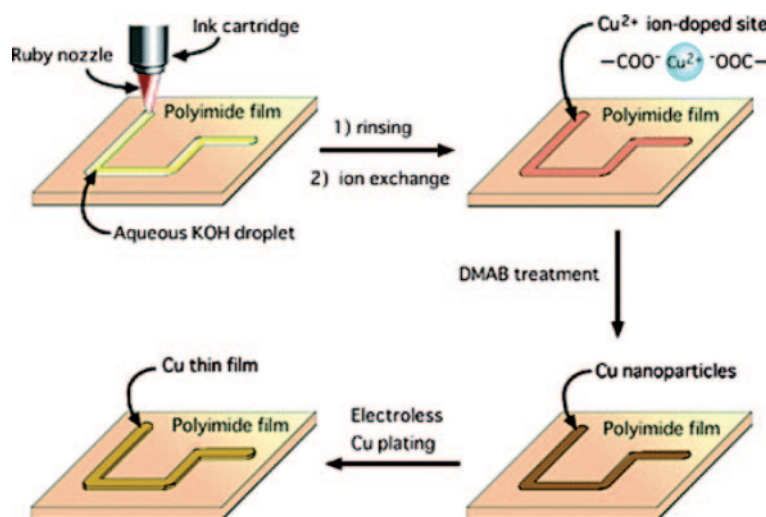


Fig. 12. – Copper patterning of a polyimide surface through ion exchange, chemical reduction with diluted dimethylamine borane (DMAB) followed by electroless Cu plating (reprinted with permission from ref. [215]).

deposition catalyzed by a seed silver nanoparticles layer was very recently reported [213]. The presence of Sn/Pd colloids onto the polyimide surface acts as catalyst for the following copper or nickel [214] deposition, while electroless copper patterns can also be produced as schematized in fig. 11 using alkanethiol self-assembly monolayer (SAM) on silver particles, introduced by ion exchange followed by thermal re-imidization, as a simple photoresist [209].

Akamatsu and co-workers [215] demonstrated the possible production by ion exchange techniques of resist-free, surface site-selective copper patterning for applications in microelectronics. The process, shown in fig. 12, allows to achieve printed circuits with satisfactory spatial resolution (see fig. 13c) and conductivity (see fig. 13b).

Direct surface metallization has been achieved on the polyimide surface by exploiting ion exchange of potassium polyamate or polyamic acid with Pd^{2+} [216] or Ag^+ ions [217–219] followed by thermal or UV-induced reduction of metal. For both Pd or Ag coatings, enhanced polyimide surface conductivity and reflectivity have been obtained. Lin and co-workers [219] very recently prepared silvered PI films with an optimum reflectivity of 81/80% and surface resistance of $0.5/0.4 \Omega/\text{sq}$ on the upside/underside.

In fig. 14 the synthetic route for silvered polyimide prepared in [218] is shown; ion exchange is performed by immersion in a solution of AgNO_3 of a $40 \mu\text{m}$ film of polyamic acid, obtained by condensation of two monomers, BTDA (benzophenonetetracarboxylic dianhydride) and ODA (oxydianiline), in a highly polar, non-protic solvent (DMAC, dimethylacetamide). After prolonged ion exchange treatment, the film is heated up to 300°C , causing reduction of silver ions and, meanwhile, the occurrence of imidization with ring closure. As the treatment time at high temperature is prolonged, agglomeration and coalescence of silver clusters take place, thereby forming an almost continuous reflective and conductive layer on both sides of the film, as evidenced by SEM images reported in fig. 15.

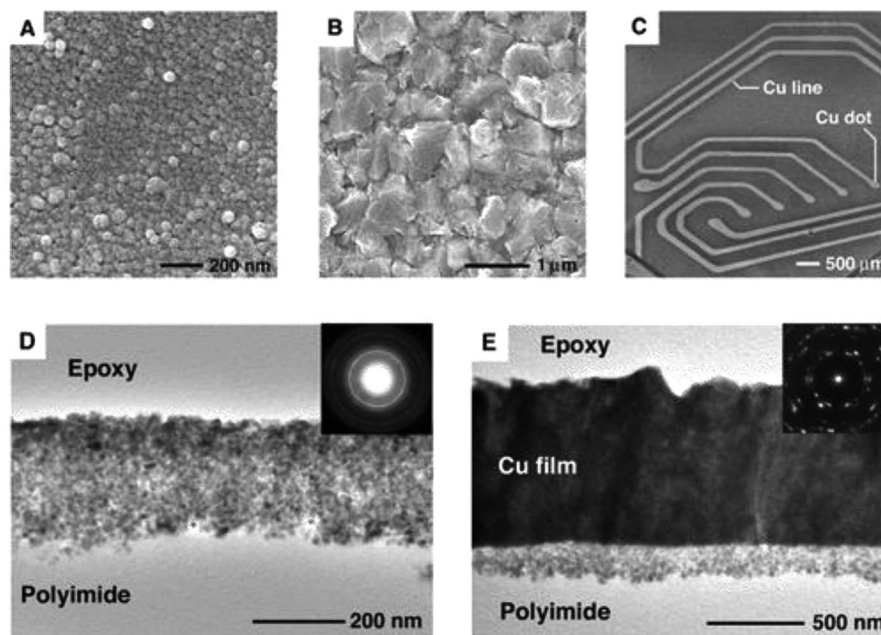


Fig. 13. – Copper circuit pattern on a polyimide surface. (A) FESEM image of metallized area obtained after DMAB treatment. (B) FESEM image of the copper film after electroless deposition on the film shown in A. (C) FESEM image of copper circuit patterns consisting of lines of $200\text{ }\mu\text{m}$ width and dots of $300\text{ }\mu\text{m}$ diameter. (D) Cross-sectional TEM image of the film shown in (A). (E) Cross-sectional TEM image of the film shown in (B). Reprinted with permission from ref. [215].

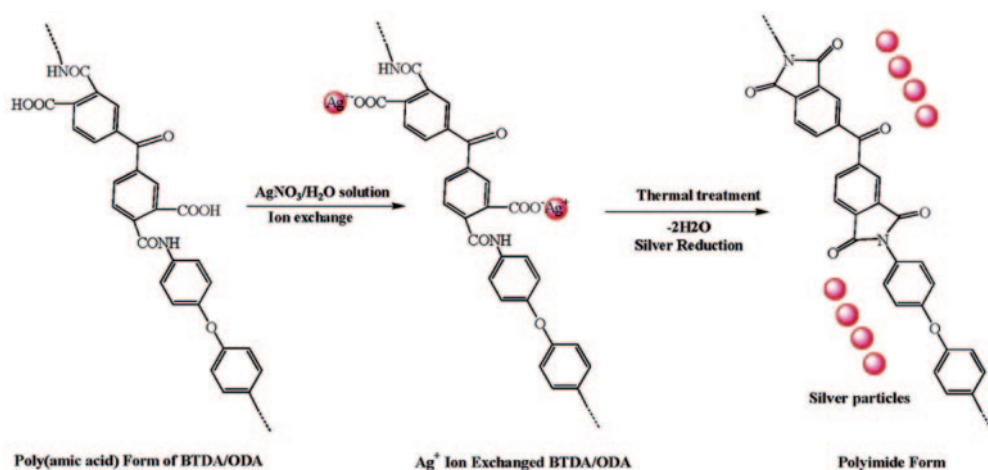


Fig. 14. – Reaction scheme for double-surface-silvered polyimide films via direct ion exchange (reprinted from ref. [218]).

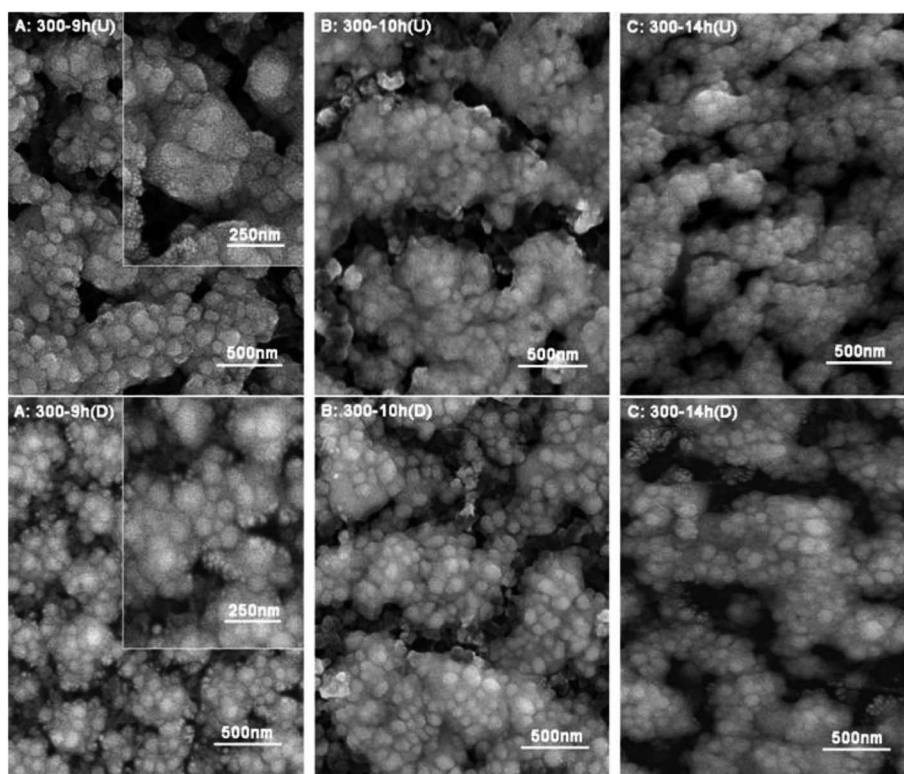


Fig. 15. – FE-SEM micrographs for the two surfaces of the composite Ag/polyimide films after ion exchange and curing at 300 °C for (A) 9 h, (B) 10 h, and (C) 14 h; U: underside of film; D: upside of film (reprinted from ref. [218]).

Embedding metal nanoclusters in polyimide is not only useful for the adhesion-promoting interlayer for further metal coating, but is *per se* a topic of exceptional interest for a tremendously wide range of applications. Noble metal nanoparticles (NP) such as gold, silver and copper display unique optical properties; for a ten of nanometers scale, a strong absorption band appears in the visible range, as a result of the collective oscillation of electrons in the conduction band, a phenomenon called Localized Surface Plasmon Resonance (LSPR) [220]. The position and shape of the plasmon resonance absorption peak depend both on the particle dimensions and on the dielectric function of the surrounding medium. For this reason, NP containing systems can be sensitive to the chemical environment, owing to the shifts or intensity changes, resulting from the adsorption or permeation of analyte molecules with an extent depending on the analyte species and concentration and on the kind of interaction with the host medium. This behaviour has been largely exploited in a wide variety of sensing applications, such as gas and vapours sensing, molecular biorecognition, oligonucleotides optical labelling [221-223].

Polymer/metallic nanoparticles composites combine the advantages of the nanoparticles, as for optical, catalytic, mechanical, electrical and antimicrobial—in case of silver NP—features with the processability of the polymers: these materials open a new gateway in developing nanocomposite systems with improved performances [224-226]. Emphasis has been given to the entrapment of metal nanoclusters in polymer matrices

by physical methods, such as ion implantation [221, 227, 228] and co-sputtering plasma polymerization [229].

Synthesis through chemical processes can also be pursued [230]: the introduction of nanoparticles can be achieved *in situ*, *i.e.* blending polymeric precursors with metal-organic compounds, followed by metal reduction by heat or UV radiation or chemical reducing agents. By *ex situ* methods, metal nanoparticles are firstly prepared as colloidal suspensions, then added to the polymer solution. These methods can induce degradation of the polymer matrix as a side effect of nanoparticles formation: in the *in situ* route towards polymer nanocomposites the dispersion of nanoclusters within the host to produce a homogeneous material can be difficult, whereas the *ex situ* method can cause polymer structure deterioration resulting from bonds cleavage induced either by UV irradiation, high temperature or chemical interactions.

As for polyimides, to date different synthetic techniques have been developed in order to obtain polyimide-metal nanocomposite thin films. Long ago, the pioneering work of Angelo laid the first stone on the fabrication of metal-polyimide nanocomposites [231] through *in situ* decomposition of organometallic complexes to entrap metal or metal oxides in a polymer matrix. More recently, multilayered metal-nanoparticle/polyimide composite thin films have been fabricated through layer-by-layer assembly [232] and by impregnation of polyimide thin films in organometallic compounds of Pt and Pd dissolved in supercritical CO₂ [233]. This latter method, besides being costly and time consuming, can be successfully applied only using metal organic precursors able to withstand the harsh conditions of fabrication as for pressure and temperature without decomposition.

A thorough investigation on the entrapment of metal nanoclusters in polyimide by ion exchange has been carried out by the group of Akamatsu and co-workers, who performed the synthesis of metal clusters, namely Ni, Pd, Cu, Ag, into commercially available polyimides obtained by ion exchange technique followed by annealing in reducing atmosphere [234-237]. The method relies on a surface treatment of the bare polyimide films in either KOH or NaOH to produce the surface-modified polyimide layer (potassium polyamate) with ion exchange capability, as described in sect. 8'2. Thereafter, metal ions are introduced through ion exchange with the bound potassium (or sodium, if the used alkali is NaOH) ions, then annealing in reducing atmosphere leads to re-imidization of the polyimide and metal nanoparticles precipitation. Unfortunately, commercial aromatic polyimides, like Kapton® or Upilex®, are not optically transparent in the whole visible range, owing to CTCs presence as described in sect. 8'1. Therefore, the optical properties of nanoclusters with typical LSPR absorption in the VIS range, such as Ag, which are known to exhibit a peak in the range of 370–480 nm depending on the matrix and particle size, cannot be determined. As previously proved by transmittance spectra of fig. 9, fluorinated polyimides exhibit improved optical transparency in the visible range with respect to commercial polyimides [238]. This feature allows the production of metal-polyimide nanocomposites in which the optical properties of nanoclusters are fully exploited. Recently, the preparation of silver nanoclusters into two transparent fluorinated polyimides (6FDA-DAD and BPDA-3F, see fig. 8 for the chemical structures) by an ion exchange technique has been performed [239, 240], aiming at a deeper comprehension of the Ag-doping mechanism and at the realization of a volatile organic compounds (VOCs) optical sensors. As for the ion exchange mechanism, Rutherford Backscattering spectrometry (RBS) and Infrared spectroscopy (FTIR) have been adopted to fully determine the structural evolution during alkali attack and re-imidization, amount of potassium and silver loading after ion exchange, stoichiometry and kinetics of the reaction.

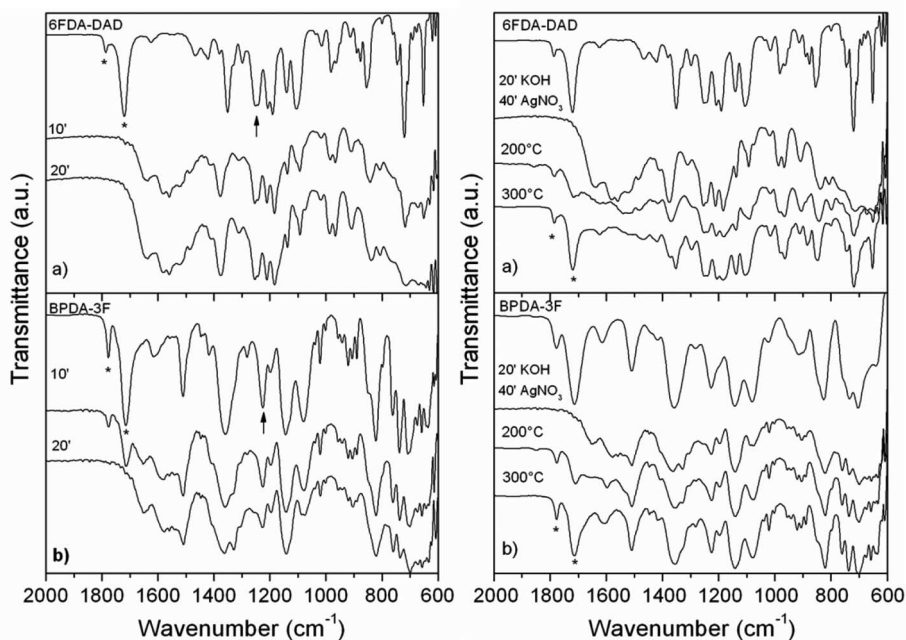


Fig. 16. – Left: ATR FT-IR spectra for polyimide films of 6FDA-DAD (a) and BPDA-3F (b) recorded after different treatment times in KOH 5M at 45 °C. The Imide I spectral features are marked with an asterisk, whereas the reference band is indicated by the arrow. Right: ATR FT-IR spectra of 6FDA-DAD (a) and BPDA-3F (b) recorded after treatment in KOH for 20 min followed by ion exchange in AgNO_3 solution and annealed at 200 and 300 °C. The Imide I spectral features are marked with an asterisk. Reprinted with permission from ref. [239].

In the FTIR spectra of fig. 16, the ring opening reaction caused by the nucleophilic attack of OH^- on the carbonyl carbon of the imide moiety is clearly evidenced by the decrease of both the signals at about 1780 and 1720 cm^{-1} , assigned to the symmetric and asymmetric stretching of the carbonyl groups of the imide ring (imide I). A different modification speed of the two polyimides can be also observed and can be explained on the basis of different reactivity of the imide carbonyl groups. In 6FDA-DAD the electron withdrawal action of the hexafluoroisopropylidene unit of the dianhydride portion promotes the reaction between the carbonyl carbon with the strong nucleophile OH^- , whereas the presence of the weak electron donating biphenyl group in BPDA-3F adversely affects the carbonyl reactivity towards nucleophile attack. Nevertheless, in both cases, the reaction speed is almost constant, though different for each polyimide type, at least within the time range used for the experiments. As expected, the mechanism appears as governed by a pseudo first-order kinetics with a linear increase of polyamate groups with treatment time in KOH. Annealing the samples in reduced atmosphere (H_2 12% - Ar) at temperatures lower than or near the glass transition temperature (T_g) values (200 °C, 300 °C and 350 °C) for 30 min allowed thermal reduction of silver and particles aggregation, as observed by the appearance of the LSPR absorbance peak at about 430 nm.

A correlation between the silver nanocrystals size and the interchain distance in the two polyimides has been investigated by analysing the nanocomposites by TEM and XRD, but size differences were found to be negligible, as can be observed in fig. 17.

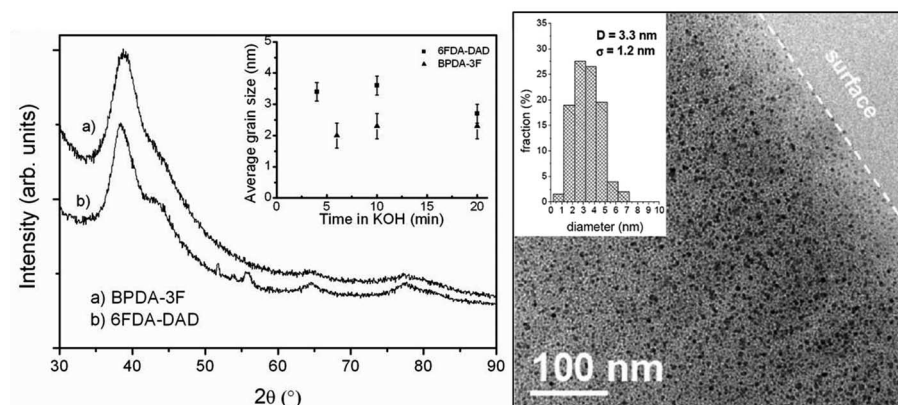


Fig. 17. – Left: XRD spectra of Ag/PI nanocomposites (in the inset graph the grain size as derived from Scherrer's formula is shown). Right: TEM image of Ag nanoclusters in 6FDA-DAD (10 min in KOH, 20 min in AgNO_3 solution, annealing at 300°C in reducing atmosphere). In the inset, the size histogram is reported [239].

Exploitation of the optical features of these nanocomposites for VOCs sensing has been reported [240]. Optical changes in the LSPR peak position and intensity have been monitored upon exposure to different analytes and good sensor response has been observed as for response and recovery times and sensitivity, as shown by the kinetic curves of fig. 18.

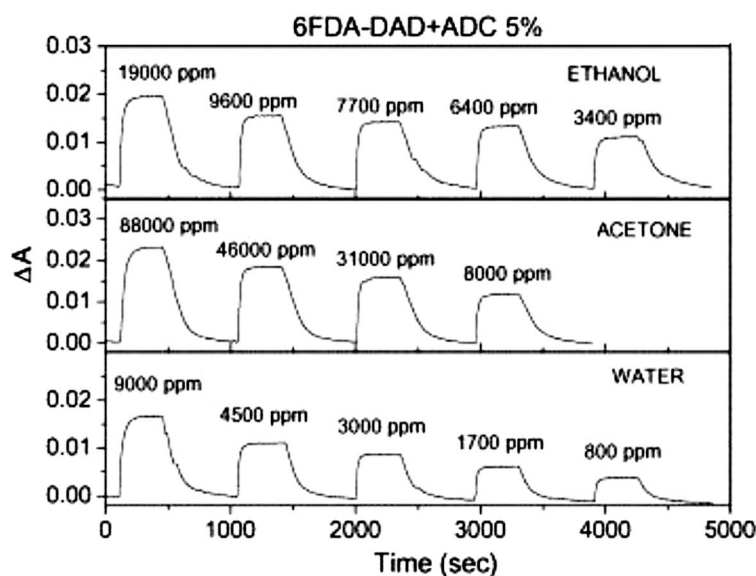


Fig. 18. – Dynamic absorbance variation in the presence of vapours of ethanol, acetone and water at different concentrations of the 6FDA-DAD thin-film doped with Ag nanoclusters. Reprinted with permission from ref. [240].

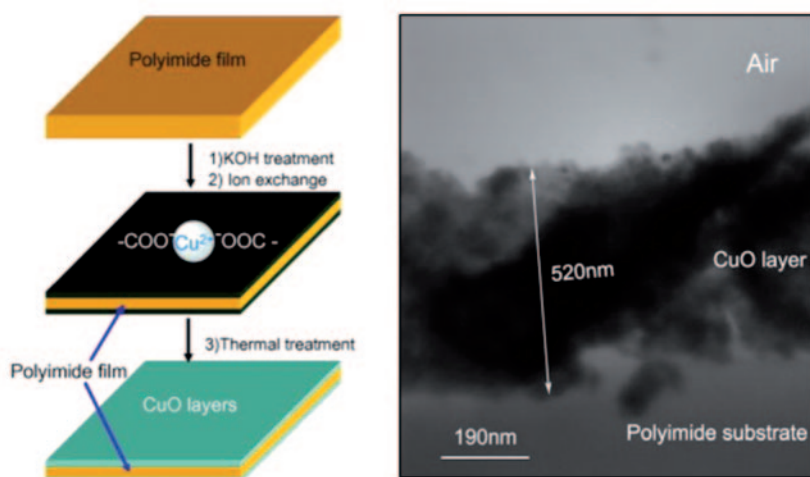


Fig. 19. – Left: scheme of the preparation of two-side CuO-doped PI. Right: TEM image showing CuO nanoparticles entrapped in PI after thermal treatment in air at 350 °C for 7 h. Reprinted with permission from part d) of ref. [241].

More recently, metal oxide nanoparticles, such as ZnO, CuO, NiO, Co_3O_4 , with potential applications in optoelectronics, light-emitting diodes, photonics, catalysts, magnetic materials, anodes for lithium batteries and solid state gas sensors and several other emerging fields of activity have been synthesized into polyimides by ion exchange [241]. ZnO and CuO nanoparticles uniformly dispersed in the surface-modified layers of PI films have been formed by air annealing, as shown in fig. 19, whereas a prolonged treatment in refluxing urea solution was adopted after Ni^{2+} ion exchange to assure formation of nickel hydroxide prior to thermal treatment for NiO precipitation.

In the field of photovoltaics, in order to broaden the applications of Dye Sensitized Solar Cells (DSSC), some research groups have made efforts in searching for other substrates for flexible DSSC applications. Recently, the technique of alkali modification followed by ion exchange with Pt/Ni based salts has been adopted [242] and successfully produced flexible polyimide substrate with Pt/Ni counter-electrode for building up polyimide based, flexible DSSC, with optimal photovoltaic conversion efficiency (higher than 7%).

The vast landscape of bare metal or metal oxide nanoparticles which can be embedded within polyimides by the ion exchange technique has been just roughly sketched in this overview, whose scope is simply to provide the reader with the impression of great versatility, ease of application, low level of toxicity and NP size and amount control capability of this method.

The applicability has been thoroughly demonstrated for a huge variety of topics of interest spanning from biological and biomedical research to materials science, physics, chemistry and environmental research. Nevertheless, the technique is so mature and within reach also of non-chemistry experts that further widening of the method to emerging fields such as organic photovoltaics, where plasmon-induced efficiency boosting in solar cells has recently come to light, is undoubtedly possible [243].

9. – Ion exchangers

An ion exchanger consists of a matrix “framework”, carrying a positive or negative fixed electric charge, which is compensated by mobile counter ions of opposite sign, as a consequence of the electroneutrality requirement. The ion exchange is essentially a diffusional process, but is also related to chemical reaction kinetics. Usually the ion exchangers are selective, taking up some counter ions in preference to others. Ion exchange systems, as inorganic clays, zeolites and specialized polymeric systems present important industrial applications such as pharmaceutical, food processing, chemical synthesis, biomedical, hydrometallurgy, water treatment, chromatography for selective removal of unwanted ions [244]. Conventional ion exchange beads consist of three-dimensional covalent networks to which ion exchange groups are bonded, that provide either cationic or anionic exchange sites.

Moreover the beads materials are susceptible to fracture and breakage due to osmotic shock, that require to maintain the beads in wet condition. The application requests, in particular of rapid regeneration and selectivity to the removal of specific ions such as heavy metals, supported the concept of developing ion exchange materials in the form of fibers about 40 years ago by J. Economy, L. C. Wohrer and F. J. Frechette, who obtained the IR-100 Award in 1972. In particular, coating glass fiber substrates with a polymeric ion exchange resin was developed, with a large number of advantages over the conventional ion-exchanged beads [245].

Cationic exchange fibers were prepared by coating glass fiber substrates with a polystyrene/divinylbenzene oligomer, curing and sulfonating. Sodium chloridesalines in contact with the resin are rapidly depleted of ions by proton exchange from the resin. The usefulness of the ion exchange process is connected to the ability to regenerate quickly the material. Another important application concerns the removal of toxic inorganics as heavy metals (Pb, Hg) present in wastewaters from the plating industries. Polymeric anionic exchange fibers are produced functionalizing with trimethylamine. A particular interest was the removal of arsenate ions from drinking water. Experimental study showed the removal of arsenic and lead to below part per billion levels, indicating the potential for removal of other toxic contaminants. A significant study has been presented [246] on the mechanisms of drug binding and drug release from cation exchange fibers *in vitro* under equilibrium conditions. The achievement of controlled drug delivery and stability of drugs during storage is strictly connected to transdermal drug delivery applications.

“Mother Nature” introduced the ion exchange process in the proton uptake and metal adsorption on algae. Algae are used as biosorbent for recovery of metals from industrial solutions, either to sequester toxic metals or to recover precious metals. The first applications are reported in the literature in 1986 [247,248]. In some cases concentration factors $> 10^6$ have been reported. Such adsorption of metals may lead to use the algae of algae in wastewater treatment or as monitoring organisms for Cu and Hg in estuaries [249].

The metal ion binding mechanism in biosorption may involve different processes such as complexation, coordination, electrostatic attraction or microprecipitation, whereby ion exchange plays a major role in the binding of metal ions by algal biomass. Therefore the use of ion exchange reaction constants instead of Langmuir- or Freundlich-type sorption isotherm relationship has been recommended to describe the process [250]. The later simplistic isotherms do not account for the effect of the back reaction of the displaced ions. This makes them unsuitable for modeling sorption processes where there is still significant binding of the displaced ions, *e.g.* at low concentrations of the sorbed metal. Ion exchange

model in multi-metal system involving multiple binding sites has been developed. The sorption performance prediction by the resulting model has been examined for a specific typical multi-metal system [251].

Sorption of metals on algae cell walls has been treated in terms of two models: adsorption in which the metal becomes bonded to one of several unoccupied sites with no additional changes at that site (adsorption isotherms, Langmuir, Freundlich), and ion exchange in which a metal displaces another ion in the sorption process. The widespread use of Langmuir isotherm in sorption studies is largely due to longstanding applications and its convenience in determining maximum sorption capacity. Such model presents some problems because it frequently does not really apply to the actual chemical process involved. Indeed the sorption of metal ions is accompanied by displacement of other cations. The ion exchange approach appears therefore more consistent with the chemical system.

Adsorption of Pb and Al on a particular algae, known as *Rhizoclonium*, under acidic conditions (metals present as ions) releases H, Ca and Mg ions from the algae, while adsorption of Pb, Cd and Al, where these metal ions are present in very low concentrations in equilibrium with solid hydroxide, also releases OH from the hydroxides [252]. Adsorption of Sr on *Vaucheria* released an equivalent amount of Ca and Mg, indicating that the metal by alkali and alkaline-earth metals is an ion exchange phenomenon based on electrostatic interactions. Release of protons when Cu was adsorbed demonstrated additional covalent bonding for this transition metal [253].

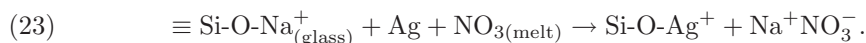
Several advantages of peat moss as sorbants to remove heavy metals from water present several advantages due its abundance, low cost, high metal capacity. The passive removal of toxic heavy metals such as Cd, Cu, Zn, Pb, Cr, Hg requires that the substrate display high metal uptake and selectivity as well a suitable mechanical properties for applied remediation scenarios. Brown algae appear to be the most effective substrates, due to cell wall constituents, such as alginate and fucoidan, responsible for an enhanced performance among biomaterials. Dead cells sorb more metal than live cells, but also pretreatments, as with CaCl_2 , enhance metal absorption of activation of algal biomass. The exchange process rate depends upon many factors as acid concentration, particle size, agitation and time of acid residence and is governed by a film-diffusion mechanism and the diffusion of Ca across a thin liquid film enveloping the particle was suggested to be the rate-determining step [254].

Sodium, along with calcium, magnesium, iron, and aluminum cation is attached on the exchange sites on the lignite structure and can be replaced by ion exchange with other elements. The metal uptake by kraft lignin, through ion exchange process, occurs by displacement of protons or bound metals. A study of equilibrium constants governing the displacement processes supported the interpretation of up-taking in terms of ion exchange and not a simple adsorption. It was found that the stoichiometry for Sr and Cd uptake by Ca-loaded lignin is 1 mol of metal for 1 mol of Ca released. This observation for metals of very different binding strengths is difficult to rationalize with the biotic ligand model as generally applied but is in complete agreement with an ion exchange process. Binding strengths to lignin, which contains only oxygen ligands, follow the order $\text{Pb} > \text{Cu} > \text{Zn} > \text{Cd} > \text{Ca} > \text{Sr}$ (strongest to weakest). For proton displacement, only more tightly bound metals such as Pb, Cu, Zn, and Cd can compete with protons for anion-binding sites at low pH, but at high pH, uptake of Ca, Sr, and Li can occur. Significant study for industrial applications has been performed on lignin obtained as a byproduct from the paper and pulp industry in the form of powder and beads (obtained upon polymerization), used for the removal of trivalent and hexavalent chromium, lead

and zinc from aqueous solutions. It appears that oxygen functionalities such as phenols present in lignin may serve as sites for the metal-cation exchange. The metal bearing lignin upon a wash with 10% sulfuric acid results in elution of copious amounts of metal. The dissolved metals in a concentrated form in the acid can be precipitated by addition of alkali resulting in metal recovery.

10. – Positron Annihilation Spectrometry (PAS) applied to ion exchange process study

PAS is a well-established spectroscopy tool for studying open volume defects in all types of materials: metals, semiconductors, polymers [255-257]. It is possible to depth profile from few nm to several microns in the samples by tuning positron implantation energy from few eV to about 30 keV. Injected positrons slow down to thermal energy in few picoseconds, then they start a diffusive motion ending their life by annihilation with an electron of the medium. At each implantation energy, a fraction of positrons diffuses at the surface of the sample annihilating in a surface state, a fraction annihilates in a bulk state and a fraction is trapped into open volume defects where it annihilates. Positrons have high sensitivity to single vacancy, vacancy clusters and open volumes up to nano-size. They are preferentially trapped in these sites due to the missing positive charge of the ions. When the open volume reaches the sub-nanometer size, positronium (ps) can be formed, the electron-positron bound state. The positron lifetime spectra of many glasses spectra can be resolved in two components, an intermediate lived component τ_1 and a long time lived component τ_2 with intensities I_1 and I_2 . The τ_1 is ascribed to the free positron annihilation and para-positronium annihilation. The positron lifetime is inversely proportional to the local electron density in the annihilation site. As a consequence, lifetime values increase when positrons become trapped in larger open volumes. Also τ_2 is ascribed to orto-positronium pick-off annihilation that is dependent on the size of free volume defects. The PAS study of mixed-cation glasses revealed free-volume anomalies in it through the measurements of the positron lifetime parameters [258]. The results were consistent with a mechanism of ion transport involving cooperation between hops of unlike cations resulting in increased disturbance of the glass network. M.H Shaa-ban *et al.* [259] investigated, by using PAS technique, the structure of soda-lime silica glass (15% Na₂O) and silver states in ion-exchanged/heat treated glass. The ion exchange process was performed by using silver nitrate as a source of silver. Heat treatment temperatures were in the range 400-600 °C for 1 h. In the ion exchange process only Na⁺ diffused from the glass surface to silver nitrate melt whereby their sites are replaced by only Ag ions. Most of the silver introduced in the glass was in the ionic state considering that the ion exchange temperature was below the glass transition temperature. The ionic radius of Ag⁺ is 1.26 Å to be compared the Na⁺ radius of 0.96 Å, The reaction is represented by the equation



The ion exchange process decreases the size of free volume and consequently increases its electron density in spite of the fact that the Ag⁺ ion has more electron negativity than the replaced Na⁺ ion. Consequently the introduction of silver ions decreases the τ_2 and I_2 values before heat treatment. Largest values of τ_2 and I_2 are found for the un-exchanged glass, characterized by free volume densities few orders of magnitude higher than those in pure crystalline solids. After heat treatment a sharp decrease of τ_2 and mean

free volume, V_f , is observed, which reflects the decrease of defect size. Heat treatment at 600 °C enhanced the structural relaxation and the agglomeration of the free volume defects, as revealed by the increase of τ_2 and V_f due to the clustering of Ag^0 atoms which form metallic Ag nanoparticles.

During an ion exchange experiment glass strengthen is developed as a result of the high surface compression due to the stuffing of the invading alkali ion into smaller host alkali ion site and the non-relaxation of this compression. The process of “ion exchange strengthening or chemical strengthening” is described in the following section.

11. – Strengthening of glass by ion exchange

11.1. Residual stress production. – The exchange of small alkali ions such as Na^+ contained in an alkali-containing glass with larger ions such as K^+ from a molten KNO_3 bath at temperatures below the strain point of the glass is typically responsible for the creation of bi-axial residual compressive stress on the surface layers of the material. Since glass products usually break due to the excessive tension applied on a surface flaw, the introduction of a surface compression strengthens the component. This process, typically called “ion exchange strengthening” or “chemical strengthening”, was initially described by Kistler [260] and Acloque and Tochon [261] in 1962, showing how the interdiffusion of ions could increase the strength by 3-4 times. It represents one of the techniques proposed and used for strengthening glass articles that include surface etching, coating and cladding, surface crystallization and, among all, thermal tempering [262-264]; this is actually the dominating process in the commercial strengthened glass market because of its relative ease of application (and, therefore, its cost effectiveness) and its relative forgiveness to handling flaws due to mostly, for instance, the nearly 1 mm depth of compression layer in a 5 mm thick glass. Chemical strengthening is a second process from an economical point of view. Compared to the thermal tempering process, chemical strengthening is advantageous in that the developed surface compression intensity is usually much higher, there is generally no measurable geometric distortion on the surface and it can be applied to thin and complex geometry products such as tubes or containers. The slow nature of ionic diffusion, as already pointed out, however, requires very long processing times (of the order of several hours) and, hence, chemical strengthening is expensive. Nonetheless, several chemically strengthened glass products have been successfully marketed, such as aircraft cockpit windshields, transparent armor, glass substrates for harvesting solar energy, high strength-to-weight ratio glass containers, auto-injector cartridges, photocopier glass, computer disks, thin display windows in communication electronic devices (monitors, cell phones, tablets and MP3 players).

Chemically strengthened glass products are nowadays covered by international standards, probably the most important being the European norm EN 12337 regarding the chemically strengthened soda-lime silicate glass used in buildings [265] and the American standard ASTM C 1422 which concerns strengthened glass products that originate from flat glass. The former defines the minimum requirement for the chemical strengthened glass in terms of 4-point bending strength [266]; conversely, the second classifies the ion-exchanged glasses on the basis of the surface compression level and the case depth (thickness of the surface layer under residual compression).

Ion exchange processes carried out at temperatures lower than T_g , where the glass can be considered as an elastic solid, are typically associated with the substitution of larger for smaller ions in the material surface that tends to expand the glass structure. But this expansion is restricted by the glass interior and, consequently, the surface layers

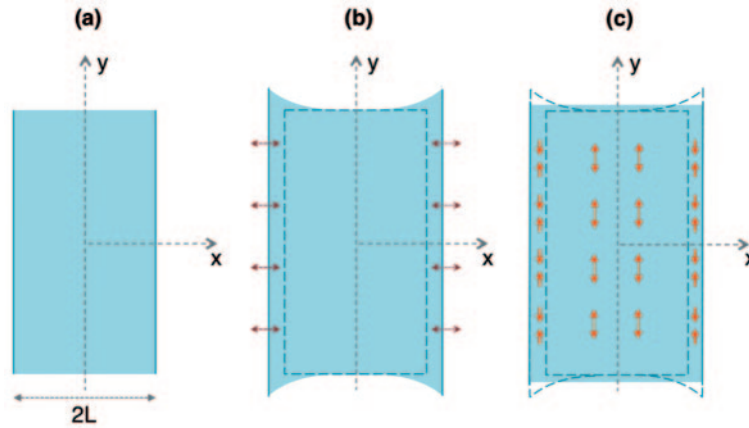


Fig. 20. – Dimensional evolution of an infinite slab during ion exchange: (a) as-received, (b) ion-exchanged and unrestrained, (c) ion-exchanged and restrained.

are placed in a state of compression and a balancing tensile stress is generated in the interior [262, 267, 268].

The compressive stresses on the glass surface are therefore generated by the “stuffing” or “crowding” effect due to the difference in size between the interdiffusing ions, at a first approximation, in the same molecular site, *i.e.* without changes in the network structure formed by the silicon-oxygen bonds in the original silicate glass [262, 267, 268].

The calculation of the developed stresses was initially proposed by Cooper and Krohn in 1969 [269] who adopted the mathematical principles of stress generation from concentration gradients proposed by Richmond *et al.* [270] for the nitridation of iron using the analogy to thermal stresses.

One can consider an infinite plate (or a plate where the thickness, $2L$, is much smaller than the other two dimensions), which has been immersed in a salt bath to promote ion exchange with smaller ions in the glass substituted by larger ions from the melt; due to the geometry of the plate the interdiffusion occurs only along the x -axis and therefore the concentration of the invading ion is a function of x , only. The dimensional changes due to ion exchange can be represented as in fig. 20, where the y -axis is therefore absolutely identical to the z one and the problem is symmetrical with respect to the y - z plane. If the expansion of the surface layers along the x and y directions was free (*i.e.* with no stresses), the resultant shape of the plate would be that shown in fig. 20b. The elongation of each portion of the material depends on the concentration, c , and a coefficient identifying the linear dilatation per exchange of a concentration unit, B , usually called linear network dilatation coefficient. Such coefficient (sometimes [267, 268] also called “Cooper coefficient” in honour of Prof. Alfred R. Cooper) that can be considered as isotropic, is given by

$$B = \frac{1}{3V} \frac{\partial V}{\partial c},$$

where V is the considered volume. As shown schematically in fig. 20, the analogy with thermal expansion effect is clear: the thermal expansion coefficient corresponds here to B , and the temperature corresponds to the concentration. On this basis, if no constrains

are present, the free strain along y for a specific x value is

$$\tilde{\varepsilon}_{yy}(x) = B c(x).$$

Nevertheless, the concentration $c(x)$ is such that free expansion of each volume element (in the present geometrical situation, thin slices parallel to the y - z plane) would make the individual volume elements different in length so that they could not be fitted together after ion exchange. Since they are constrained in the same body, stresses arise (fig. 20c). These stresses can be calculated for a perfectly elastic body as occurs if the ion exchange is carried out at a temperature well below the glass transition temperature [271]. The overall strain along y (*i.e.* the average elongation of the plate) can be determined on the basis of the average concentration

$$\bar{c} = \frac{1}{2L} \int_{-L}^{+L} c(x) dx$$

and, therefore, the average strain over the thickness $2L$ is

$$\bar{\varepsilon}_{yy}(x) = B\bar{c}.$$

The stress at any point depends on the difference in the concentration between that point and the average value; therefore

$$(24) \quad \sigma_{yy}(x) = \sigma_{zz}(x) = \frac{EB}{1-\nu} [\bar{c} - c(x)] = \frac{EB}{1-\nu} \left[\frac{1}{2L} \int_{-L}^{+L} c(x) dx - c(x) \right],$$

where E is Young's modulus and ν Poisson's ratio of the material, assumed as invariant with c .

The form of eq. (24) ensures that the residual stresses are self-equilibrated and also shows that the case depth is the depth where the actual concentration equals the average one, which is readily obtained from the relative increase of weight, the glass thickness and the atomic mass of the exchanged ions. Equation (24) also allows to point out that the case depth is always smaller than the diffusion depth. As long as the exchanged depth is smaller than L , the stress in the core scales with c and therefore increases with the square root of the exchanging time [272].

According to the elastic "stuffing" theory previously described, the ion exchange process could produce a surface stress given by

$$(25) \quad \sigma_{yy}(L) = \sigma_{zz}(L) = \frac{\Delta V}{V} \frac{E}{3(1-\nu)},$$

where $\Delta V/V$ is the specific volume expansion caused by the substitution of the small ion with the bigger one. It should be recognized that the stack of mixed-alkali glass layer structures in the ion exchange region does not resemble the glass structures of the corresponding as-melted compositions; they are in reality stuffed derivatives of the parent glass structure. Therefore, the comparison of the molar volume of glasses with different composition or the volume change associated to the elastic stuffing of interstitial sites are only a rough approximation of the real effect. In any case, the maximum compressive stress that can be estimated is always in excess to 1 GPa [262-264, 267].

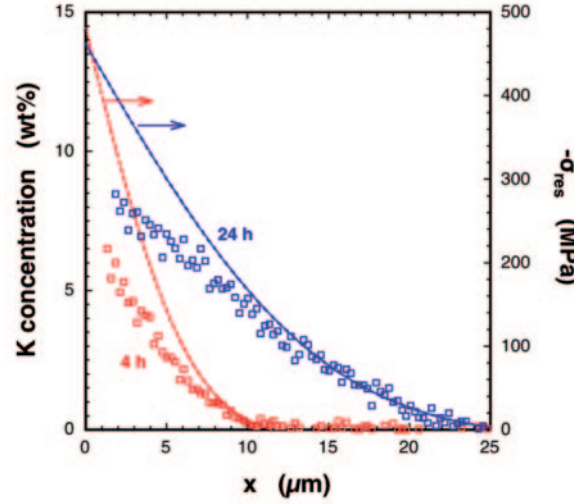


Fig. 21. – Potassium concentration profile and residual stress ($\sigma_{\text{res}} = \sigma_{yy}(x) = \sigma_{zz}(x)$) after ion exchange of soda-lime silicate float glass in KNO_3 at 450°C for 4 h and 24 h.

An example of the stress generated after ion exchange process, correlated to the concentration profile is shown in fig. 21. In the specific case, the stress $\sigma_{yy}(x) = \sigma_{zz}(x)$ was determined by a procedure based on the measurement of the tempered sample curvature deriving from progressively etching one of its surfaces [273]; conversely the potassium concentration was measured by EDXS micro-probe.

There is a general agreement between the stress and the concentration profile of the invading larger ion; nevertheless, there is also an evident discrepancy between the theoretical compression and the actual compressive stress induced by ion exchange. As from the numerous previous works, it is possible to point out that the maximum residual stress achievable by ion exchange is around 2–5 times smaller than the calculated one [262–264, 267].

If the ion exchange process is carried out at temperatures not far from T_g , viscoelastic or plastic effects can reduce the generated stresses. As proposed in [272], a stress relaxation function, ψ , strongly depending on temperature (*i.e.* with the same activation energy of glass viscosity) can be considered for quantifying the viscoelastic effect and eq. (25) can be modified as

$$(26) \quad \sigma_{yy}(x, t) = \sigma_{zz}(x, t) = \frac{EB}{1 - \nu} \int_0^t \psi(t - \tau) \frac{\partial}{\partial \tau} [\bar{c}(\tau) - c(x, \tau)] d\tau,$$

where $\bar{c}(\tau)$ and $c(x, \tau)$ are the average concentration and punctual concentration at time τ , respectively, and t the processing time.

From eq. (26) one can obtain a relationship between the compressive stress on the surface ($x = L$), $\sigma_{yy}(x, t)$, and the stress in the core, $\sigma_{yy}(0, t)$,

$$(27) \quad \sigma_{yy}(L, t) = \sigma_{zz}(L, t) = \sigma_{yy}(0, t) - \frac{EB}{1 - \nu} \psi(t).$$

For a higher temperature process, ψ tends to zero more quickly, so do the stress components. Actually, the core stress still increases with $t^{1/2}$ at the beginning, so that the viscoelastic relaxation mainly decreases the surface compression, quickly for a high processing temperature. On the other hand, the viscoelastic relaxation does not directly affect the concentration profile, nor much the case depth, which can be increased in a shorter time by using higher temperature. Therefore, there exists a best process temperature compromise, for a given maximal allowable processing time, which depends on whether the emphasis is on the maximum surface compression or on a large case depth. Stress profile measurements show that the surface compression stress decreases with increasing exchange time, but the longer the processing time the greater the case depth. There is therefore a time at which the modulus of rupture of a glass reaches a maximum for each specific ion exchange temperature. The higher the temperature, the shorter the time needed to produce this maximum strength, although in general higher temperatures lower such maximum strength.

In addition to viscoelastic relaxation, structural relaxation can also take place for long processing times and/or relatively high processing temperature. Early suggestions of network plasticity were proposed by various authors [274, 278].

As proposed by Ingram *et al.* [279], near T_g , the cations may themselves lead to stress reduction by causing local bonds to break and rearrange through cation-induced relaxation of network (“CIRON”), eventually leading to a fully relaxed structure which is statistically identical to an as-melted glass. More recently, Varshneya has emphasized [280, 281] the need for incorporating network hydrostatic yield strength and shear yield strength into the estimation of the coefficient B . When a glass interstice is stuffed by an invading ion, the network elastically expands at first building up stresses due to enforcement of compatibility criteria and then plastically deforms (under the effect of deviatoric stress field that preserves the volume but changes the shape) or accommodates the size disparity by bond bending/stretching (due to a hydrostatic stress field that promotes densification without changing the overall shape). During plastic flow or densification the stress generated under pure elastic regime is partially reduced. It is clear that if the aim is to generate high magnitude beneficial compression stresses, the yield strength or plastic limit must be high.

Hydrostatic yielding of glass network leads to “irreversible” densification. Recently, techniques such as Raman spectroscopy and molecular dynamic simulation have been useful [282–284] in identifying the atomic mechanisms that lead to the densification. It has been shown that plastic behaviour varies differently with network topology, *i.e.* with the coordination number corresponding to each site. Similar plastic deformation effects have been also observed during indentation of silicate glasses [285–287]. By extending this argument to the ion exchange phenomenon, it may be concluded that well-connected glasses deform elastically to a greater strain before approaching the yield strength. Likewise, lower alkali content glasses are expected to have a higher plastic limit due to the lesser non-bridging oxygen fraction.

Because the yielding in glass occurs around an exchange site only, it would be important to know the relationship of this local stress field with the overall mesoscopic stress production. Buried within this relationship is the nonlinear elasticity (second-, third-, and fourth-order Young’s moduli as discussed by Kurkjian *et al.* [288]) and irrecoverable (plastic) strain.

An additional effect is related to the stress decrease occurring upon cooling the glass from the ion exchange temperature down to room temperature. Tyagi and Varshneya [289] have shown that this reduction is around 20–25% in alkali aluminosilicate

glasses and is associated to a high thermal contraction coefficient of the exchanged layers, which is about 40% larger than the thermal expansion coefficient of the bulk glass although the origin of such discrepancy is not known. In some cases, the surface compression has been shown to change to a mild tension upon progressive ion exchange, even at the exchange temperature, thus not being related to the relatively high thermal expansion coefficient of the surface layers [290, 291]. Also in this case the origin of this large decrease in stress magnitude and eventual changeover is not clear. Finally, one should remind that although the exchanged layers are “stuffed” derivatives of the bulk glass (eventually viscously or plastically modified or rearranged) and their structure differs from as-melted glass of the same composition, sensible variations in the thermal expansion coefficient can complicate further the evolution of stresses upon cooling.

11.2. Strengthening. – The strength of a brittle material like silicate glass is not an intrinsic property, with a definitely recognized value. It is well known that failure stress is a statistical quantity, absolutely correlated in glass to the statistical distribution of surface flaws that originate during processing, handling and service. Such “defects” can be associated to geometrical, compositional or structural discontinuities as initially postulated by Griffith [292]; more practically, they correspond to inclusions, second phases, pores or scratches. Especially in silicate glasses, whose hardness is lower than that of most atmospheric powders, whose surface contact/impact thermal shock resistance is relatively small, and whose chemical resistance to humidity is relatively weak, one can imagine the surface absolutely dotted with flaws, randomly variable in length, depth and orientation. The presence of a defect on the glass surface has the effect to amplify the applied tensile loads, thus causing “critical” conditions for failure at its tip. There is therefore a combined effect of the external load and the defect on the final resistance of the material. Conventionally, according to fracture mechanics principles, such combined effect is expressed by the stress intensity factor defined as [293]

$$(28) \quad K_a = \varphi \sigma_a \sqrt{C},$$

where σ_a is a characteristic external applied stress, C a characteristic length of the defect and φ the shape factor, which includes the geometry of the component and the defect itself. If one considers a two-dimensional configuration (probably the most typical in fracture mechanics for its simplicity) constituted by a semi-infinite plate with a surface flaw orthogonal to the external applied stress (fig. 22a), the shape factor is equal to $1.12\sqrt{\pi} \approx 2$; in a more realistic situation (but more complicated from a calculation point of view) (fig. 22b), a small semi-circular surface crack (radius = C) is subjected to pure tension (stress = σ_a) only, the crack plane being orthogonal to the applied stress, the shape factor is equal to $2.24/\sqrt{\pi} \approx 1.26$.

Failure conditions (or critical conditions) occur when the stress intensity factor equals the material resistance or better the critical stress intensity factor, K_C , which is an intrinsic property of the material, *i.e.*,

$$(29) \quad K_a = K_C.$$

For most silicate glasses, $K_C = 0.7\text{--}0.8 \text{ MPa m}^{0.5}$ [293–296], which, compared to other materials, is a relatively small number; this justifies the relatively limited mechanical resistance of as-produced glass when typical micrometric-size surface defects are present.

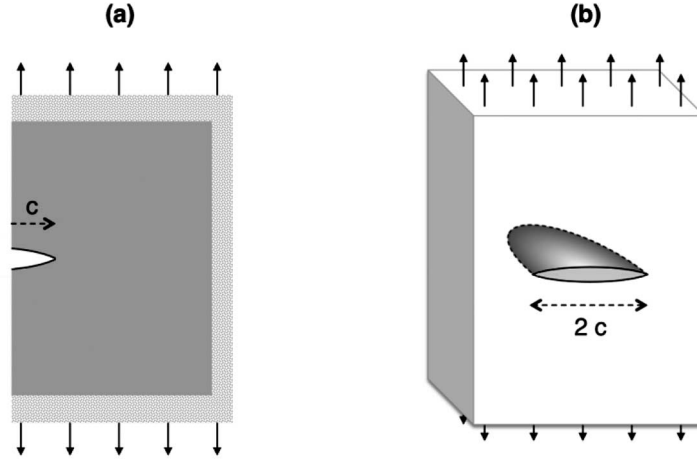


Fig. 22. – Typical crack configurations used in fracture mechanics.

On the basis of eqs. (28) and (29), the strength of a glass article depends on the size of the “critical” defect as

$$(30) \quad \sigma_F = \frac{K_C}{\Psi \sqrt{C}}.$$

Therefore, due to the statistical distribution of the “critical” defects in a body, also the strength has to be considered statistically distributed. This makes the strength also depend on the size of the component and the loading configuration, the probability of finding a “big” flaw being higher if the surface under tension is larger.

The formation of residual stresses by ion exchange is responsible for additional “external” loads acting on the material, whose effect can be positive (reinforcement) or negative (weakening) depending whether the residual stresses are compressive or tensile. To follow the typical fracture mechanics approach previously defined, the mathematical calculation of the residual stress effect requires the definition of the associated stress intensity factor. Only for simplifying the calculations but without losing in generality, the configuration corresponding to a surface crack in the semi-infinite plate (fig. 22) can be considered. In this case, the stress intensity factor corresponding to the residual stresses is [273]

$$(31) \quad K_{\text{res}} = \frac{1.12}{\sqrt{\pi c}} \int_0^c \sigma_{\text{res}}(x) g(x) dx,$$

where

$$(32) \quad g(x) = \frac{2c}{\sqrt{c^2 - x^2}}$$

is the Green’s function and $\sigma_{\text{res}}(x)$ the residual stress (equal to previously defined $\sigma_{yy}(x) = \sigma_{zz}(x)$). For the situation in which the residual stresses are compressive,

there is the possibility of crack-faces contact. In this case, eq. (31) is not valid anymore [297]; however, when the two stress intensity factors are superposed, *i.e.* external tensile stresses are applied in the region of the crack, one can easily demonstrate that the crack faces are completely open and eq. (31) can be used in the superposition. For this situation, the total stress intensity factor is

$$(33) \quad K_{\text{tot}} = K_a + K_{\text{res}}.$$

Under the condition for crack growth ($K_{\text{tot}} = K_c$), eq. (33) becomes

$$(34) \quad K_c = K_a + K_{\text{res}}.$$

In fracture mechanics, one usually includes stresses in the crack “driving force”; however, as noted previously [298, 299], it is sometimes useful to consider residual stresses as part of the crack resistance. An apparent fracture toughness [298] can be therefore defined as

$$(35) \quad \bar{K}_c = K_c - K_{\text{res}}(c) = K_c \frac{1.12}{\sqrt{\pi c}} \int_0^c \sigma_{\text{res}}(x) g(x) dx.$$

It is clear that in this case failure occurs when

$$(36) \quad K_a = \bar{K}_c.$$

Thus, for compressive residual stresses, an apparent toughening effect is developed, because K_{res} is negative and $\bar{K}_c$ is larger than K_c ; on the basis of eqs. (30) and (31), this means that the strength of the glass is higher for similar flaw size.

On the basis of eq. (35), the toughening effect strictly depends on the residual stress profile. This can be quite variable depending on the processing conditions and on the glass, as previously pointed out, although, in several cases, an “erfc” function can be used to describe the σ_{res} profile. In such case, anyway the mathematics to be used for the determination of the apparent fracture toughness, $\bar{K}_c$, is quite complicated. Without losing in generality and with the aim to point out the fundamental effects of the residual stress field on toughening and strength, the simplifying assumption that the residual compression linearly decreases from a maximum value on the surface, σ_{res}^0 , down to zero at the depth, d_0 , (the case depth) can be considered. Such assumption clearly is not very far from typical residual stress profiles generated by ion exchange. Depending on the value of σ_{res}^0 and d_0 , different $\bar{K}_c$ are obtained. Some examples are shown in fig. 23, where the apparent fracture toughness curve corresponding to various σ_{res}^0 and d_0 values and calculated by eq. (35) are reported and compared to the applied stress intensity factor K_a defined by eq. (28) for a semi-infinite plate with a surface flaw (fig. 23a). The assumption is made here that the glass has a fracture toughness equal to $K_C = 0.7 \text{ MPa m}^{0.5}$. Only the surface region where the residual compressive stresses are effective is considered, this being the layer where toughening is produced. It is clear that cracks larger than d_0 can be hardly reinforced and actually a weakening associated to the tensile residual stress in the inner region of the glass, though with limited intensity, shall be considered. In fig. 23 one can observe that toughening is more effective for larger σ_{res}^0 and, especially, d_0 values and this is associated to a sensible strength increase. In the specific examples, almost no toughening is produced when the maximum residual compression stress is equal to 100 MPa or even 200 MPa; for higher compressive stresses the reinforcement is

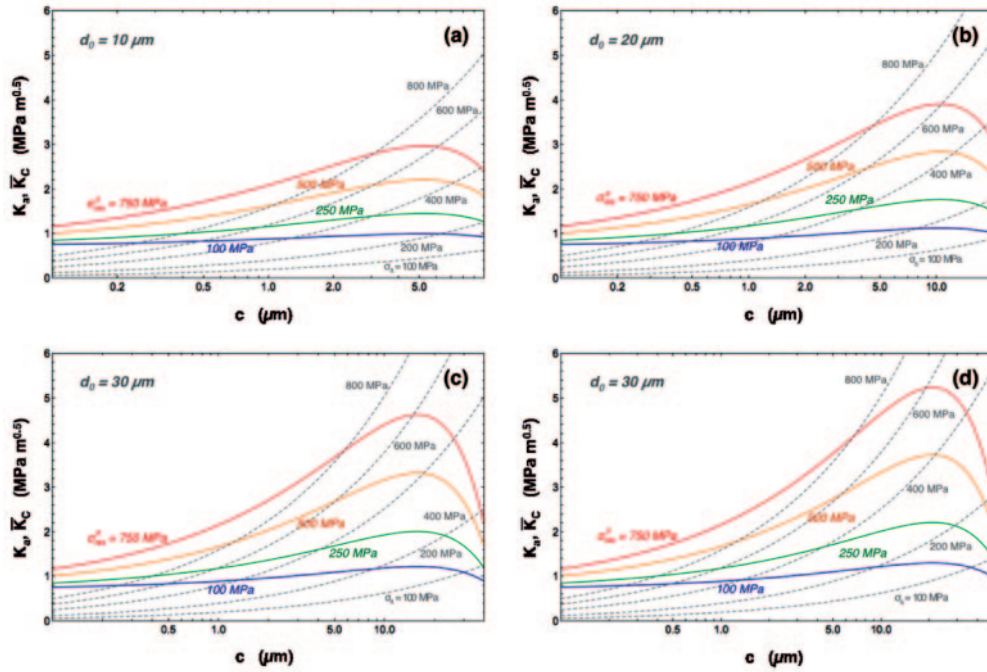


Fig. 23. – Apparent fracture toughness ($\bar{K}_C$) calculated for different σ_{res}^0 and d_0 values and applied stress intensity factor (K_a) for different applied stresses as a function of crack size.

instead clearer. The intersection points between the apparent fracture toughness and the applied stress intensity factor in fig. 23 are used to evaluate the allowable crack size for obtaining a specific final strength as reported in fig. 24. It is possible to observe that very high failure stresses (in excess to 800 MPa) are possible only for small defects. Relatively lower strength in the range 400–600 MPa, still being absolutely remarkable for glass, can be obtained also when quite deep flaws (20–35 μm) are present in the material, provided that well-developed compressive stress (with $d_0 = 40 \mu\text{m}$ and $\sigma_{\text{res}}^0 = 500$ –750 MPa) are produced. The general guideline that can be advanced on the basis of the results in fig. 25 for the ion exchange strengthening of glass is that the surface residual compressive stress should be maximized for obtaining high failure stress values and shield most of the defects on the surface. One can therefore easily understand the importance of knowing the fundamental mechanisms on the basis of the residual stress development previously described and of the research still being carried out on this matter.

In practice, typical surface cracks on industrial as-produced float glass are equivalent to surface notches with depth of the order of 10 μm ; according to fig. 24, a case depth of the same size provides significant reinforcement if the maximum residual compressive stress is larger than 200–300 MPa, as typically obtained by conventional ion exchange process. Nevertheless, such a relatively small case depth may not be of great help for applications where glass is subjected to contact damage. Several works have shown that chemical tempering accounts for higher contact damage resistance with respect to thermal tempered or annealed glass, this implying also better abrasion resistance [272,

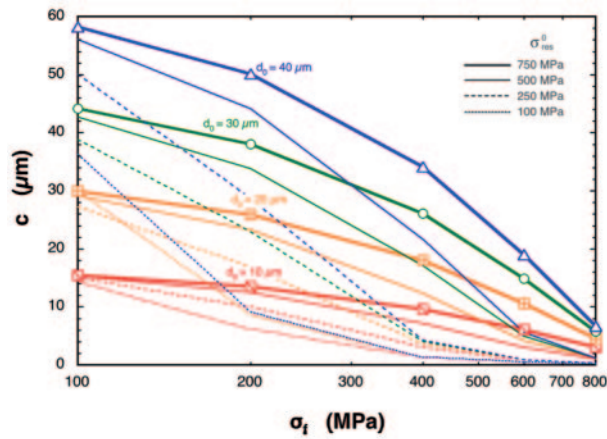


Fig. 24. – Allowable maximum crack size for obtaining a specific final strength as calculated on the basis of fig. 23.

293,300,301]. The effect is anyhow limited up to a certain threshold contact load and beyond that limit the damage depth is larger than the case depth and the strength of the chemically tempered glass falls down to the level of the annealed glass. Therefore, there are two opposite effects that have to be taken into account in the optimization of the chemical strengthening process. If one aims to achieve a very high glass strength, the surface compression has to be maximized; conversely, if the glass surface can be subjected to severe contact damage during service, then a large case depth has to be favoured and this generally comes at the price of a lower undamaged strength because of a lower compressive stress on the surface.

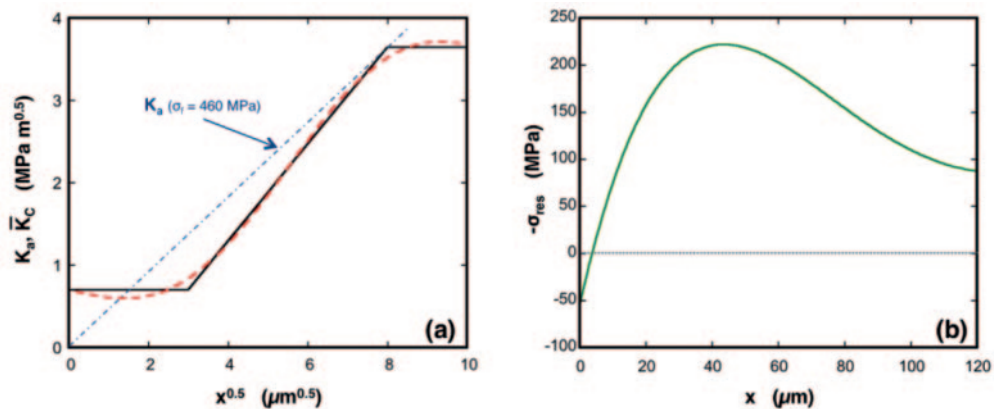


Fig. 25. – (a) Apparent fracture toughness that allows stable growth of surface defects; dot-dashed line corresponds to the applied stress intensity factor associated to “final” strength of 460 MPa; dashed curve is the polynomial approximation used for the evaluation [307] of the residual stress profile (b) which guarantees the specific $\bar{K}_c$ curve.

An additional concept can be recalled here regarding the contact damage resistance of glass. In some previous works the concept of “brittleness” has been advanced, commonly defined as the ration between hardness, H , and fracture toughness, and often used to represent an increased susceptibility to crack initiation [302-305]. The main idea is that it is the crack initiation (*i.e.* brittleness) that triggers the reduction of the strength of glass by a few orders of magnitude when compared to their theoretical strength; and several research activities are devoted to try to decrease the brittleness of glass by specific modifications of composition and processing. In a recent work, Koike *et al.* [300] showed that in chemically tempered soda-lime float glass the crack initiation load during Vickers indentation increases with the surface compressive stress although the modification of the glass structure (by substitution of K for Na) makes them more brittle. More generally, there is a relationship between the possible irreversible deformations (plastic and viscous) of the glass network with contact damage resistance. Therefore, the optimization of the glass network for the creation of higher residual stresses by ion exchange can be connected to the aim for producing glass with better contact damage resistance.

11'3. Engineered Stress Profile (ESP) glass. – The fracture mechanics concepts previously presented for the determination of the residual stress effect on failure stress have been extended in previous works [299, 301, 306, 307] to the analysis, design and production of glasses with high mechanical reliability, *i.e.* characterized by more deterministic strength values, as is typical for other materials such as metal alloys or plastics. As previously stated, failure in glass occurs when $K_a = \bar{K}_c$, and by using a definition usual for thermodynamics, the phenomenon is typically unstable in the sense that when the applied stress intensity factor reaches the critical value for the specific crack size, failure starts and propagates irreversibly to the material since $K_a > \bar{K}_c$ for any crack length larger than the initial one (see fig. 23). By using a mathematical notation, the unstable fracture occurs when the following two conditions are satisfied:

$$(37a) \quad K_a = \bar{K}_c,$$

$$(37b) \quad \frac{dK_a}{dc} > \frac{d\bar{K}_c}{dc}.$$

There are anyway some situations where failure occurs in a stable fashion, *i.e.*, although $K_a = \bar{K}_c$, crack can propagate only if the applied stress intensity factor is increased further, otherwise the crack arrests. This happens if

$$(38a) \quad K_a = \bar{K}_c,$$

$$(38b) \quad \frac{dK_a}{dc} < \frac{d\bar{K}_c}{dc}.$$

These latter are typically defined as the conditions for the stabilization of the crack growth [293].

Several efforts have been done in the past to identify techniques that can be used to stabilize or arrest crack growth. It has been shown that either a proper modification of microstructure [293] or the introduction of a particular residual stress profile can generate a so called “rising R - or T -curve behaviour” [299, 301, 306, 307], *i.e.* the

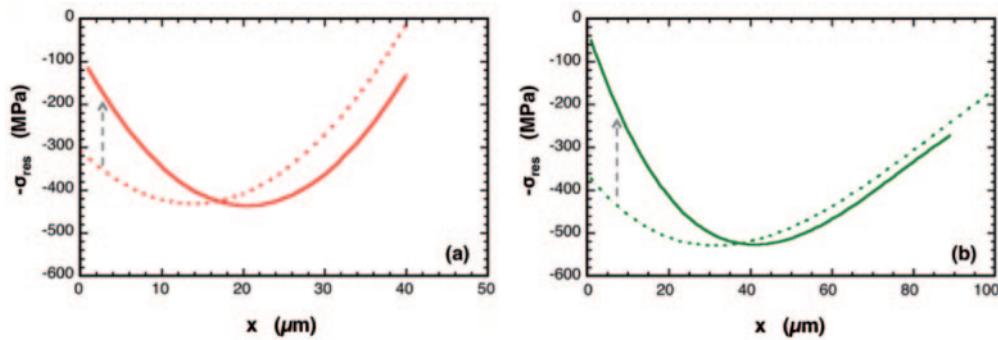


Fig. 26. – Examples of residual stress profiles obtained by double ion exchange process on soda-alumina silicate glass [308]; the arrows indicate the effects of the second treatment in $KNO_3/NaNO_3$ bath.

flaw encounters an increasing fracture resistance during its propagation; in other words, the fracture toughness increases more rapidly than the applied stress intensity factor. It has been demonstrated that materials that exhibit unique *R*-curve behaviour can show reduced strength scatter and, more importantly, the strength is insensitive to flaw size [298,308]. The idea of designing particular forms of the residual-stress distribution is particularly attractive in glasses, because there is no microstructure to control and thus, microstructural toughening mechanisms are not available. The main thrust in the theoretical calculations was to move the maximum in the surface compression (more rigorously, a stress minimum) away from the surface and then determine whether crack growth could be stabilized [299].

The ideal residual-stress profile that allows surface crack arrest can be defined by using fracture mechanics approach. A model has been developed that allows the evaluation of the residual stress profile associated to a pre-defined *T*-curve [299]. In this way the strength of the component can be defined *a priori* and the residual stress field designed once the flaw size distribution and the fracture toughness of the pristine material are known. The fundamental idea is to promote a stable growth of surface defects before final catastrophic failure. In this way, regardless the initial defect size, a constant failure stress can be obtained (fig. 25a). One important feature is that the residual stress profile must possess a compressive maximum at a certain depth from the surface (fig. 25b); in addition, the gradient of the residual stress profile between the surface and the maximum compression must be sufficiently high in order to promote the stable growth of surface flaws [299].

The conceptual idea was reduced to practice by using a double ion exchange on silicate glasses. The versatility of the ion exchange process was pointed out in such a work to create a residual stress with tailored profile. The glasses subjected to such two-step ion exchange treatment were therefore named ESP (Engineered Stress Profile) glass as the production process is strictly connected to the engineering of the residual stress profile [301,306,307]. Both soda-lime silicate and sodium aluminosilicate glasses were considered. Processing procedure consists of a first relatively long step (from 24 to 120 h) in KNO_3 bath where K^+ ions are exchanged for Na^+ ions in the glass and a second shorter step (30 min), where a fraction of the potassium ions are exchanged back for sodium ions in a $KNO_3/NaNO_3$ mixed bath. This second process is necessary to move the maximum

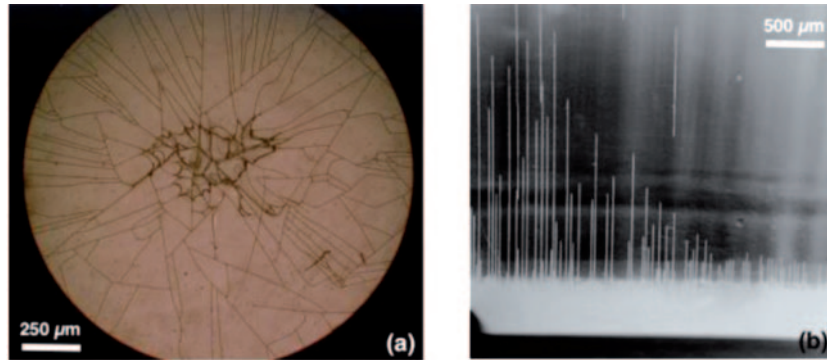


Fig. 27. – Stable propagation of cracks on the tensile surface of (a) disk subjected to biaxial flexure and (b) bar in four-point bending.

compression to a region below the external surface as shown in fig. 26. This design and creation of the stress profile leads to some important attributes in the mechanical behaviour of the glass. As with the traditional ion exchange processes, the compressive stresses still give rise to significant strengthening [301,306,307]. The difference is that the stress gradient in ESP glass is such that cracks can be arrested without causing failure even in normally unstable stress fields, *i.e.* those associated with applied stress intensity factors that increase as cracks grow. This effect leads to a reduction in strength variability (lower than 5%) and to multiple cracking prior to the final failure that “warns” of the impending failure (fig. 27) [309]. Moreover, the strength of ESP glass is less sensitive to surface damage (fig. 28) and to sub-critical crack growth [301].

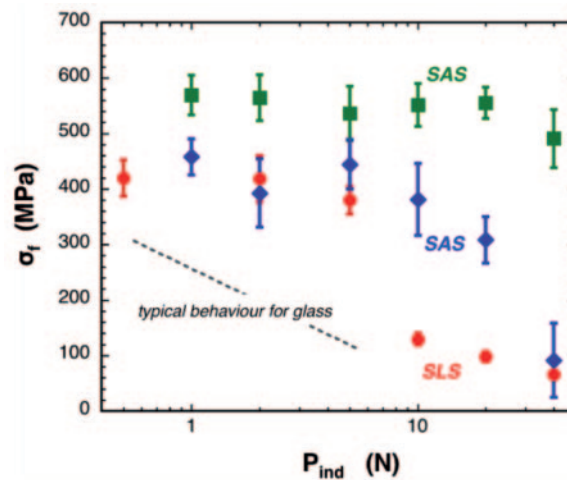


Fig. 28. – Strength as a function of the Vickers indentation load for soda-lime (SLS) and soda-alumina (SAS) silicate glasses [308].

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